

Biotechnology
MapLight Therapeutics, Inc. (MPLT)

RATING BUY	PRICE TARGET \$37.00	PRICE \$24.14	52-WEEK RANGE \$12.24 - \$24.15
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No Time for Side Effects: A Safer and More Convenient Alternative to Cobenfy

We are initiating coverage of MPLT with a Buy rating and \$37 PT based on a SOTP-based DCF. Our thesis is fourfold: (1) The muscarinic class is likely to be disruptive with application across multiple indications including schizophrenia and AD psychosis. (2) ML-007C-MA should be differentiated from Cobenfy in schizophrenia on tolerability and convenience supporting \$1.5B in sales by 2035. (3) ML-007C-MA could be even more differentiated from Cobenfy in AD Psychosis on tolerability and convenience (BID vs TID), supporting sales of \$2.4B by 2035. (4) ML-004 is an early call option with Ph 2 IRIS data expected in 3Q26.

ML-007C-MA's Ph 2 ZEPHYR data in schizophrenia (3Q26) and Cobenfy's ADEPT-2 Ph 3 readout in 2026 should meaningfully de-risk this story along with the call option on ML-004 in 3Q26.

KEY POINTS

[1] The muscarinic class is likely to be disruptive with improved safety and efficacy over SoC as evidenced by Cobenfy, and have application across multiple indications including schizophrenia and AD Psychosis

- In schizophrenia Cobenfy demonstrated a better clinical profile compared to SoC antipsychotic treatments including a higher ES of 0.60-0.61 vs 0.3–0.5 for SoC and no safety concerns around weight gain, EPS, and cardiovascular concerns
- The muscarinic class also has potential to work in other neuropsychiatric indications with preclinical studies demonstrating that targeting M1 and M4, muscarinic receptors can impact psychosis and cognition.
- KarXT's Ph 3 ADEPT-2 results expected in 2026 should de-risk the class in AD psychosis.

[2] Lead asset, ML-007C-MA is likely to be differentiated from Cobenfy in schizophrenia on tolerability and convenience both of which have been concerns in the RW, with at least comparable efficacy, supporting \$1.5B in sales by 2035.

KEY DATA

Market Cap (MM)	\$1,024.4
Price [04/07/2026]	\$24.14
52-Week Range	\$12.24 - \$24.15
Shares Outstanding	42.44
Avg. Daily Volume	215,120.3

ESTIMATES

FY (Dec)	2025A	2026E	2027E
EPS			
Q1	-	(1.61)E	-
Q2	-	(1.57)E	-
Q3	(37.18)A	(1.47)E	-
Q4	(2.47)A	(1.10)E	-
Year	(18.56)A	(5.75)E	-
Growth		NM	
P/E	NM	NM	
Rev. (MM)(\$)			
Year	0.0A	0.0E	-

- ML-007C-MA demonstrated a better safety and tolerability compared to KarXT in both healthy adults and elderly volunteers (Figure 23, Figure 31). In healthy adults at the target Ph 2 doses (210/3mg BID and 330/6mg QD) ML-007C-MA was differentiated from Cobenfy on
 - **(i) vomiting**, a procholinergic AE, with 0% rate vs 27.8% for Cobenfy's 100/20mg dose, a key driver of discontinuations in the RW, and
 - **(ii) anticholinergic AEs**, with 0% rate for the target Ph 2 doses vs 16.7% (100/20 mg BID) and 66.7% (125/40 mg BID) for Cobenfy,
 - **(iii) urinary retention**, for which Cobenfy has a warning in its label (7% severe, with 3.5% requiring hospitalization), but was not seen in ML-007C-MA's study.
 - It had higher rates of hyperhidrosis (17-33%) with none reported for Cobenfy; this is less of a concern.
- ML-007C-MA has a meaningful convenience advantage with **no fasting** requirement, **minimal titration** (one day vs up to weeks for Cobenfy in RW) and **potentially less frequent dosing** of QD vs BID for Cobenfy.

(Continued inside)

At a Glance

OUR INVESTMENT THESIS

(1) The muscarinic class is likely to be disruptive with application across multiple indications including schizophrenia and AD psychosis. (2) ML-007C-MA should be differentiated from Cobenfy in schizophrenia on tolerability and convenience supporting \$1.5B in sales by 2035. (3) ML-007C-MA could be even more differentiated from Cobenfy in AD Psychosis on tolerability and convenience (BID vs TID), supporting sales of \$2.4B by 2035. (4) ML-004 is an early call option with Ph 2 IRIS data expected in 3Q26.

BULL CASE ASSUMPTIONS

- ML-007C-MA's PoS in Schizophrenia is 55% with 4.5% market share at peak
- ML-007C-MA's PoS in AD Psychosis is 25% with ~30% market share at peak

BASE CASE ASSUMPTIONS

- ML-007C-MA's PoS in Schizophrenia is 55% with 3.5% market share at peak
- ML-007C-MA's PoS in AD Psychosis is 25% with ~22% market share at peak

BEAR CASE ASSUMPTIONS

- ML-007C-MA's PoS in Schizophrenia is 55% with 3% market share at peak
- ML-007C-MA's PoS in AD Psychosis is 0%

COMPANY DESCRIPTION

MapLight Therapeutics is a clinical-stage biopharmaceutical company focused on improving the lives of patients suffering from debilitating central nervous system disorders. The company was founded by globally recognized leaders in psychiatry and neuroscience research to address the lack of circuit-specific pharmacotherapies available for patients. MapLight's discovery platform holds the potential to fill this void by identifying neural circuits causally linked to disease and targeting those circuits for therapeutic modulation.

PRICE PERFORMANCE



KEY DATA

Market Cap (MM)	\$1,024.4
Price {04/07/2026}	\$24.14
52-Week Range	\$12.24 - \$24.15
Shares Outstanding	42.44
Avg. Daily Volume	215,120.3

- ML-007C was 10X more potent than xanomeline (M1/M4 agonist in KarXT) in preclinical studies, which could translate into at least comparable efficacy.
- The muscarinic class can achieve >\$5B in US sales in schizophrenia by 2035 with ~12% market penetration. For ML-007C-MA, we estimate sales of \$1.5B in 2035 driven by 30% share, and estimate a 55% POS.

[3] Muscarinics have potential application in several other indications including AD psychosis, where ML-007C-MA can be even more differentiated, supporting sales of \$2.4B by 2035.

- Xanomeline has previously demonstrated benefit in AD patients on both cognition and psychosis validating the potential of muscarinic agonists beyond schizophrenia.
- ML-007C-MA can be meaningfully differentiated from Cobenfy, especially on tolerability
 - In healthy elderly patients ML-007C-MA demonstrated a more tolerable profile vs KarXT on discontinuation rates (0% vs 20%), moderate AEs (18% vs 47.2%), pro-cholinergic AEs like nausea (27% vs 33%) and vomiting (9% vs 19%) and anti-cholinergic AEs such as constipation (9% vs 50%).
 - ML-007C-MA will have a convenience advantage of a BID dose vs TID for cobenfy, in addition to the other convenience advantages discussed above.
 - Its greater M1 receptor bias than Cobenfy could potentially have a greater implication on efficacy in AD psychosis given its role in cognition and memory.
- We believe the muscarinic class can achieve >\$7.5B in US sales in AD Psychosis by 2035 with >50% market penetration. For ML-007C-MA, we estimate sales of \$2.4B in 2035 driven by 22% share of the overall market, and estimate a 25% POS.

[4] We think ML-004 is an early call option with Ph 2 IRIS topline data readout expected in 3Q26, although it is not a significant part of our current valuation.

Thoughts on the stock: Two key catalysts this year that could meaningfully de-risk this story along with an early call option for the second asset, ML-004, in 3Q26

- Ph 2 ZEPHYR data in schizophrenia in 3Q26: Our base case ([Figure 1](#)) assumes that Ph 2 confirms a tolerability and convenience profile superior to Cobenfy with comparable efficacy resulting in upside in the mid-to-high 20s (65% POS). In a downside scenario ML-007C-MA has comparable tolerability to Cobenfy (25% POS, -low 30s% stock move) and in an upside scenario, it is superior also on efficacy (10% POS, +40%).
- The ADEPT-2 Ph 3 data for Cobenfy, if positive, could de-risk this class in AD psychosis.
- Maplight also has a Phase 2 VISTA study ongoing for ML-007C-MA in AD psychosis which is expected to readout in 2H27.
- Maplight will also advance a preclinical candidate into IND-enabling studies for its next-gen M1/M4 muscarinic ML-005, which has demonstrated greater potency than ML-007, has potential QD dosing and LAI formulation.

The company ended 2025 with \$453M in cash supporting operations through 2027 key upcoming data readouts.

For the upcoming Ph 2 data readout, we see three potential scenarios ([Figure 1](#)).

Figure 1 - Ph 2 Scenarios

			PoS (%)	Stock Move (%)
Scenario 1	Base	Superior on tolerability and convenience with comparable efficacy	65	+mid-to-high 20s
Scenario 2	Downside	Comparable on both efficacy and tolerability but superior on convenience	20	-low-30s
Scenario 3	Upside	Superior on tolerability, convenience, and efficacy	10	+40s

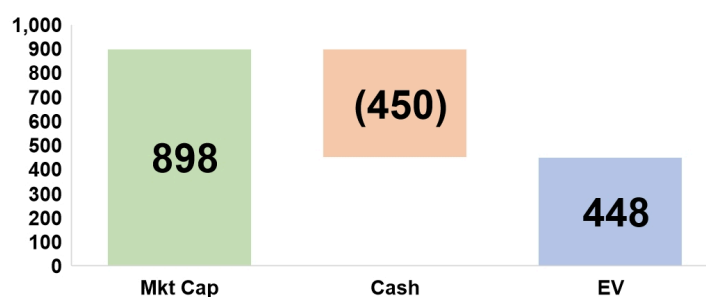
Source: Needham & Company Research

COMPANY OVERVIEW

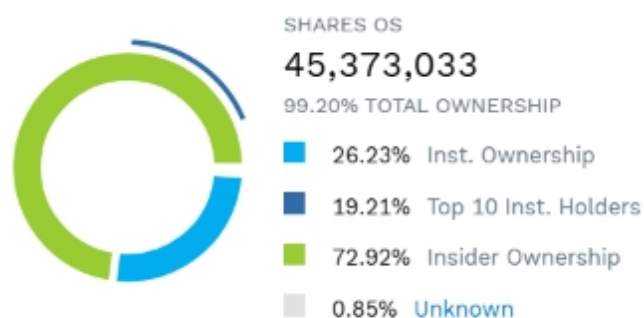
COMPANY OVERVIEW

- Maplight Therapeutics was founded in 2018 by Dr. Karl Deisseroth, Dr. Karoly Nikolich, and Dr. Chris Kroeger to develop targeted therapies for CNS disorders
- Its lead asset is ML-007C-MA in Ph 2 trials for Schizophrenia and AD Psychosis.
- As of the end of 3Q, the company had \$227.2M in cash with runway through 2027.

ENTERPRISE VALUE



OWNERSHIP STRUCTURE



TOP HOLDERS

SHORT INTEREST	FLOAT	INST. OWNERSHIP
9.5 DAYS / 11.1% FLOAT	27.1%	96.9% OF FLOAT

Source: FactSet

Rank	Type	%OS
-	Total	87.7
-	Institutions	21.8
1	T. Rowe Price Investment Managem	6.2
2	Goldman Sachs & Co. LLC (Private	2.4
3	BlackRock Fund Advisors	2.2
4	Goldman Sachs Asset Management	1.9
5	The Vanguard Group, Inc.	1.7
6	Perceptive Advisors LLC	1.2
7	Velan Capital Investment Manageme	1.2
8	Geode Capital Management LLC	0.9
9	Vestal Point Capital LP	0.8
10	Baker Bros. Advisors LP	0.8
11	Jain Global LLC	0.7
12	SSgA Funds Management, Inc.	0.5
13	Lazard Asset Management LLC	0.5
14	Stanford Management Co.	0.4
15	Citadel Advisors LLC	0.4
-	Insiders / Stakeholders	66.0
1	Catalyst4, Inc.	43.4
2	Novo Holdings A/S (Investment Com	8.1
3	Forbion Capital Partners Manageme	7.1
4	Kroeger Christopher A	3.9
5	Pivotal Bioventure Partners Investme	3.5

Source: FactSet

EXPERIENCED MANAGEMENT TEAM

MapLight Therapeutics (MPLT): Management Overview



Christopher Kroeger, M.D. *Chief Executive Officer, Founder, Board Member*

- Dr. Kroeger is the CEO, Founder, and Board Member at MapLight Therapeutics.
- Previously, he served as the CEO of Cardioxyl Pharmaceuticals, leading a team that produced three investigational new drug candidates and successfully executed six Phase 1 and Phase 2 clinical studies, culminating in the sale of the company to Bristol Myers Squibb for \$2.1 billion in total consideration
- He earned his bachelor's in Biology from Harvard University and his Doctor of Medicine from Stanford University School of Medicine. In addition, he earned his MBA from Harvard Business School



Erin Pennock Foff, M.D., PhD *Chief Medical Officer*

- Dr. Foff is MapLight's Chief Medical Officer
- Previously, she served as an executive director of clinical development at Acadia Pharmaceuticals and led the company's programs in the dementia/neurodegenerative space
- She holds a doctorate in molecular and human genetics from Baylor College of Medicine and completed her medical degree at Jefferson Medical College, residency in neurology at University of Virginia, and holds a certification from the American Board of Psychiatry and Neurology



James Lillie, Ph.D. *Chief Scientific Officer*

- Dr. Lillie is the MapLight's Chief Scientific Officer. He leads the Company's preclinical research and development efforts
- He joins MapLight from OvaScience where he served as the CSO. Previously, he served as Senior Director and, later, Vice President of In Vitro Biology at Sanofi Genzyme where he was responsible for developing a scientific strategy to create a sustainable, high value portfolio in rare diseases
- He holds a doctorate from Harvard University, and a bachelor's in German Literature from Wesleyan University



Anatol Kreitzer, PhD *Chief Discovery Officer*

- Dr. Kreitzer is MapLight's Chief Discovery Officer and leads the Company's platform development and target discovery efforts
- Prior to joining MapLight, he was a Professor of Physiology and Neurology at the University of California, San Francisco, where he led a laboratory focused on the basal ganglia, a highly-conserved brain region in vertebrates that controls motivation, decision making, and the automation of behavior
- He holds a bachelor's degree from the University of California, Berkeley, and a doctorate in Neurobiology from Harvard University

MapLight Therapeutics (MPLT): Management Overview



Vishwas Setia, MBA, Chief Financial Officer

- Vishwas Setia is MapLight's Chief Financial Officer
- Prior to joining MapLight, Mr. Setia was a Managing Director in the healthcare investment banking group at Bank of America Securities
- He holds an M.B.A. from the Kellogg School of Management at Northwestern University and a B.E. in Mechanical Engineering from Delhi College of Engineering, India



Kris Hanson, General Counsel

- Mr. Kris Hanson is MapLight's General Counsel and leads the legal and compliance functions
- Prior to joining MapLight, Mr. Hanson was a member of the executive leadership teams at PhaseBio Pharmaceuticals and Nalpropion Pharmaceuticals, where he had day-to-day responsibility, oversight, and management of the legal and compliance functions
- Mr. Hanson received a BA from Georgetown University and a JD from the UCLA School of Law



Jonathan Gillis, Chief Administrative and Accounting Officer

- Mr. Jonathan Gillis is MapLight's Chief Administrative and Accounting Officer.
- Prior to joining MapLight, he served as the Senior Vice President of Finance overseeing finance, human resources, information technology, and legal functions at OvaScience, and was instrumental in several public offerings and the ultimate merger with Millendo Therapeutics
- He holds a B.S. and a M.S. in Accounting from Babson College

COMPANY PIPELINE

COMPANY PIPELINE HAS BROAD THERAPEUTIC POTENTIAL WITH 3 ONGOING PH 2 STUDIES & 2 READOUTS IN 2026

Figure 2 - Pipeline

Program	Circuit	Indications	Preclinical	Phase 1	Phase 2	Phase 3	Anticipated Milestones
ML-007C-MA M ₁ /M ₄ agonist co-formulated with PAC	Direct and Indirect Pathways	Schizophrenia	ZEPHYR				Topline results in Q3 2026
		Alzheimer's Disease Psychosis	VISTA				Topline results in 2H 2027
ML-004 5-HT _{1B/1D} agonist	Dorsal Raphe to Nucleus Accumbens	Autism Spectrum Disorder	IRIS				Topline results in Q3 2026
ML-021 M ₄ antagonist	Direct Pathway	Parkinson's Disease					Complete IND-enabling studies in 2H 2026
ML-009 GPR52 PAM	Indirect Pathway	Hyperactivity/Impulsivity					Nominated preclinical candidate

Potential in other indications being explored

Source: Company Presentation

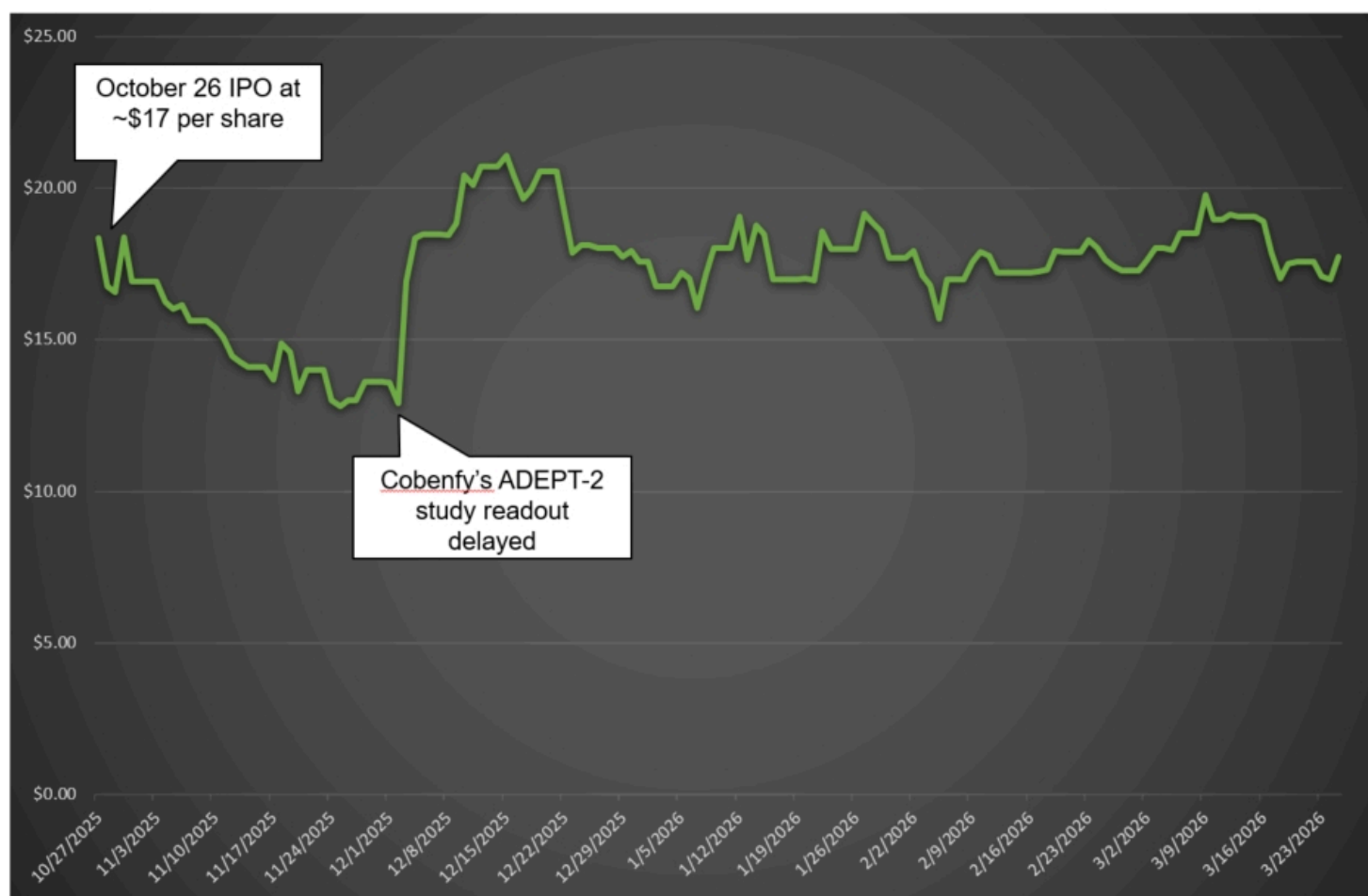
UPCOMING CATALYSTS

Name	MOA	Stage	Indication	Details	Timing
ML-007C-MA	Dual M1/M4 agonist in combination with PAC	Ph 2	Schizophrenia	Ph 2 ZEPHYR study is ongoing and expected to reach target enrollment (N=300) in April 2026. Next update is data readout.	3Q26
		Ph 2	AD Psychosis	Ph 2 VISTA study (N=300) is ongoing (Fast Track Designation)	2H27
ML-004	IR/ER oral formulation of zolmitriptan (5-HT 1B/1D agonist)	Ph 2	Autism Spectrum Disorder	Ph 2 IRIS study (N=160) has completed enrollment	3Q26
ML-021	M4 Antagonist	Preclinical	Parkinson's Disease	Complete IND-enabling studies	2H26
ML-055	Next-gen M1/M4 muscarinic agonist	Preclinical	Neuropsychiatric conditions	Nominate preclinical candidate	2026
ML-009	GPR52 PAM	Preclinical		Nominate preclinical candidate	TBD

Source: Company Reports

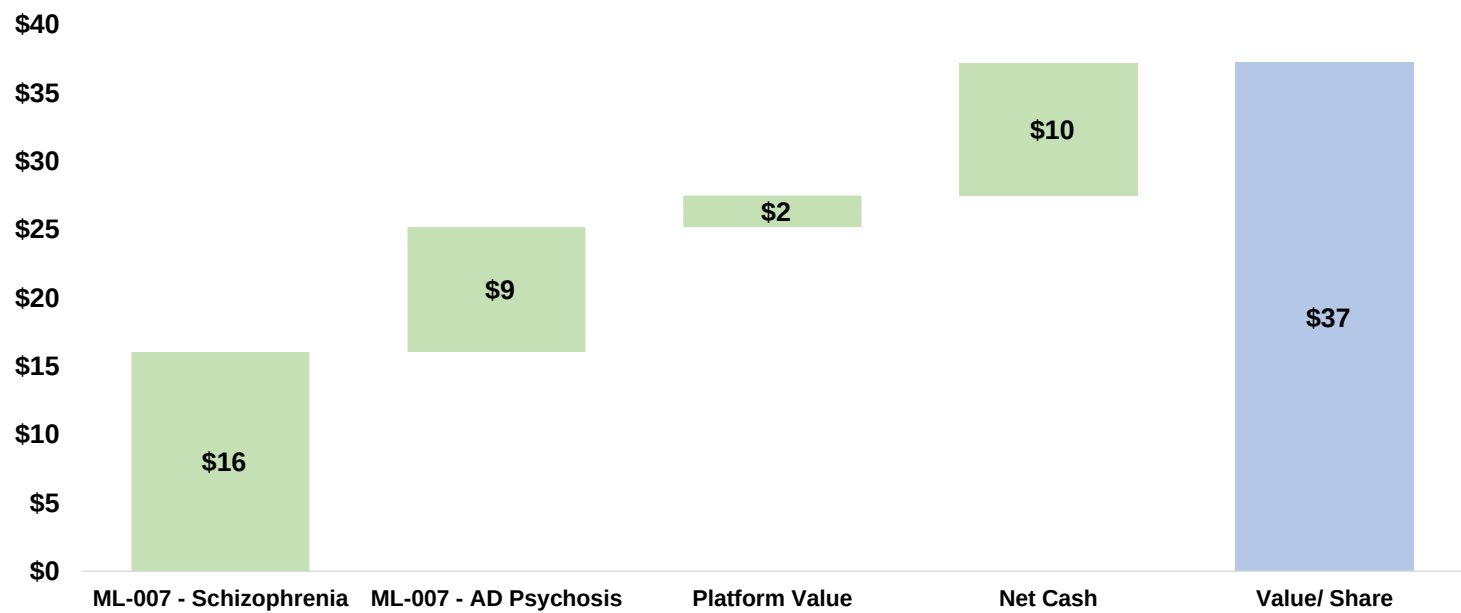
STOCK HISTORY

Maplight Therapeutics (MPLT): Stock History



Source: Team estimates; FactSet

SUM-OF-PARTS DCF VALUATION



- BMS acquired Karuna Therapeutics for KarXT at an EV of ~\$14B ([LINK](#)) in December 2023 (March 2024 deal close). At the time of acquisition, it had demonstrated positive data in schizophrenia only. While we acknowledge that at the time of acquisition, KarXT was the leading muscarinic agonist with validated data from pivotal studies, it still suggests that Maplight's potentially best-in-class asset in schizophrenia and AD psychosis should be worth more.
- Similarly, Abbvie acquired Cereval Therapeutics at an EV of ~\$8.7B ([LINK](#)) in December 2023 (August 2024 deal close). There were other assets in the pipeline, but the main valuation was based on emraclidine, M4 PAM, which was being assessed in Ph 2 studies with some PoC from Ph 1b. This is a more appropriate comparator for Maplight given that it was coming in second after Cobenfy with a potentially differentiated profile.
- Overall, the acquisitions in the muscarinic space provide us confident in our valuation of MPLT which is currently being valued at an EV of ~\$500M.

UPSIDE/ DOWNSIDE ANALYSIS

Upside/ Downside Scenarios

Key Variable	Downside Scenario	Current Stock Level	Our View	Upside Scenario	Blue Sky Scenario
	 PT: \$10 -57%	Stock Price: ~\$23	 PT: \$37 ~+60%	 PT: \$46 ~+100%	 PT: \$55 ~+140%
Schizophrenia Market Share	3.0% at peak (2035) -\$6/share	NC	3.5% at peak (2035)	NC	~4.5% +\$9/share
Schizophrenia PoS	NC	~45% -\$6/share	55%	NC	NC
AD Psychosis Market Share	NA	NC	~22% at peak (2035)	~30% +\$9/share	~30% +\$9/share
AD Psychosis PoS	0% -\$21/share	~15% -\$8/share	25%	NC	NC



Blue text shows \$/share change vs. 'Our View'

Source: Needham & Company, LLC estimates

SNAPSHOT OF INVESTMENT THESIS

MapLight Therapeutics (MPLT) – Investment Thesis Snapshot

Ticker	Needham Rating	Needham PT	Date	Close	52-week		Market Cap (M)	Average Volume (k)	Short Interest	2025 Cash/share	2025 Cash & Equiv. (M)	EV/Sales-2025E
MPLT	Buy	\$37	4/5/2026	\$12.46	Low	High	\$884	199.5	11.0%	\$13.88	\$453	NM

Investment Thesis	Needham vs Consensus Estimates (\$M, except EPS)						
- The muscarinic class, specifically ML-007C-MA's approach, has already been validated in clinic with the approval of <u>Cobenfy</u> in schizophrenia, and is likely to be disruptive with application across multiple indications including schizophrenia and AD psychosis. - We believe there is a good chance for ML-007C-MA to demonstrate a better clinical profile compared to <u>Cobenfy</u> addressing physician / patient concerns on the tolerability& dosing regimen - It demonstrated a better safety & tolerability profile compared to <u>Cobenfy</u> (KarXT) in both healthy adults and elderly volunteers - The tolerability profile, especially high rates of nausea, vomiting, and urinary retention, for <u>Cobenfy</u> have been cited as major concern by physicians - Additionally, ML-007C-MA does not have any fasting requirements and has minimal titration compared to <u>Cobenfy</u> which makes it a more convenient option as well - ML-007 is 10X more potent than <u>xanomeline</u> (M1/M4 agonist in <u>Cobenfy</u>) in preclinical studies - We estimate ML-007C-MA can grow to ~\$1.5B and ~\$2.4B in schizophrenia & AD psychosis, respectively, in the US driven by: - Improved clinical profile over <u>Cobenfy</u>, especially an improved tolerability profile & dosing regimen - Lack of approved therapies in AD psychosis - There are two big catalysts this year that could meaningfully de-risk this story along with an early call option on ML-004		2025A	2026E	2027E	2028E	2029E	2030E
	Revenues - Needham	0	0	0	0	17	128
	% Growth Rate	NM	NM	NM	NM	NM	659.1%
	Consensus		0	0	0	0	49
	Needham vs Cons.		NM	NM	NM	NM	161.0%
	EPS - Needham	-\$18.56	-\$5.58	-\$2.89	-\$3.33	-\$2.99	-\$1.41
	% Growth Rate		69.9%	48.2%	-15.1%	10.2%	52.8%
	Consensus		-\$3.95	-\$4.10	-\$4.77	-\$4.56	-\$4.67
	Needham vs Cons.		-41.4%	29.5%	30.3%	34.5%	69.8%
Key Catalysts							
	- ML-007C-MA's Ph 2 ZEPHYR study readout in schizophrenia in 3Q26 - ML-004's Ph 2 IRIS study readout in ASD in 3Q26 - ML-007C's-MA's Ph 2 VISTA study readout in AD psychosis in 2H27						
Investment Risks							
	- ML-007C-MA's safety data may not be differentiated from <u>Cobenfy's</u> - Other <u>muscarinics</u> demonstrate a better benefit risk profile - Muscarinic class does not work in AD psychosis						
Valuation							
Our \$37 PT is based on a sum-of-the-parts valuation, which forecasts cash flows for each revenue stream out to 2035E and uses an 12.5% discount rate.							

Positive data from Ph 2 ZEPHYR study should be an important de-risking event in 3Q26 along with a potential positive ADEPT-2 study for Cobenfy in AD psychosis in 2026 which should de-risk the muscarinic class in a new indication.

Source: Team estimates; FactSet

OVERVIEW

Current Schizophrenia Therapies only impact positive symptoms and have side effects resulting in poor adherence

Atypical antipsychotics are the current standard of care in schizophrenia, and they work by blocking dopamine D2 receptors, particularly in the mesolimbic pathway, which is thought to **drive the positive symptoms** of schizophrenia (hallucinations, delusions). While the atypical antipsychotics help address these positive symptoms, they also cause a **wide range of side effects** including extra pyramidal symptoms (EPS), metabolic side effects (higher in second generation drugs like olanzapine, clozapine), hyperprolactinemia, QTc prolongation and sedation.

Schizophrenia patients also suffer from negative symptoms and cognitive impairment, and the current SoC treatments do not do anything to address these symptoms. Overall, there remains a significant unmet need in this market in the real-world with 40-60% of ([LINK](#)) patients not adhering to antipsychotics treatment due to lack of efficacy and/or undesirable side effects.

Figure 3 - Schizophrenia Treatment Algorithm Summary

Line	Setting & Approach	Key Agents
First-Line	Acute inpatient or outpatient: Monotherapy with an SGA; agent selection based on patient profile, side-effect considerations, and formulation preference	Risperidone, Olanzapine, Aripiprazole, Paliperidone, Quetiapine, Lurasidone, Cariprazine, Lumateperone, Xanomeline/Trospium (Cobenfy)
Second-Line	Switch to different SGA class or formulation (oral → LAI); optimize dose; consider adjunctive therapies	Alternative SGA monotherapy; LAI formulations (Paliperidone Palmitate, Aripiprazole LAI, Risperidone LAI); augmentation strategies
Third-Line (TRS)	Inpatient or specialized outpatient: Clozapine initiation with mandatory ANC monitoring; reserved for patients failing ≥2 adequate antipsychotic trials	Clozapine (gold standard); Emerging: Evenamide (add-on, Phase III)
Adjunctive / Supportive	Across all settings: Psychosocial interventions, digital therapeutics, cognitive remediation	CBT for psychosis, supported employment, psychoeducation, CT-155 (PDT, investigational)

Source: Company reports; American Psychiatric Association. Practice Guideline for the Treatment of Patients with Schizophrenia, 3rd Edition. APA; 2020

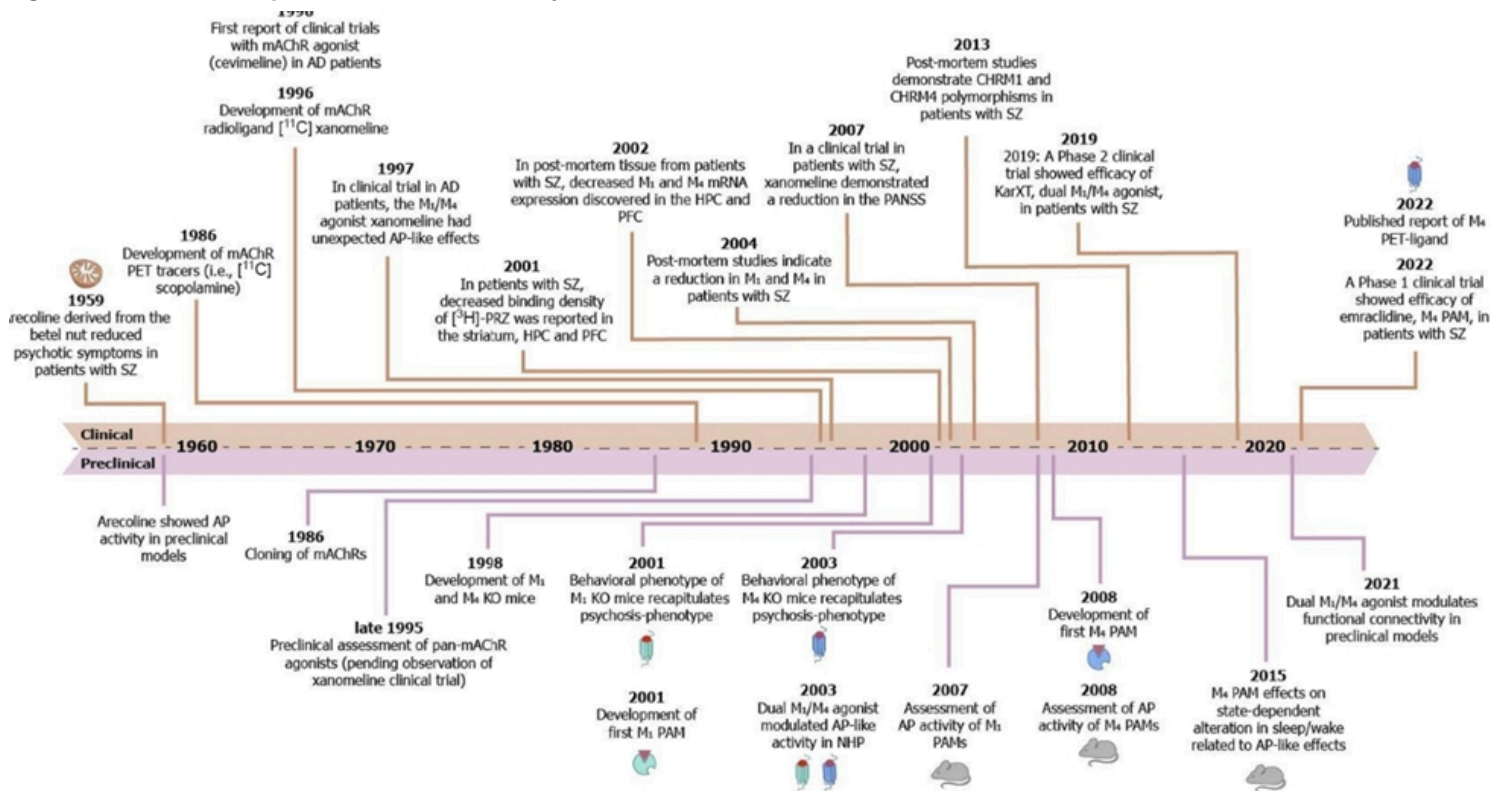
Muscarinic class has been in clinical development since the 1990s, however, the class has been previously limited due to safety and tolerability concerns arising from peripheral muscarinic activity

In our previously published deep dive note on the muscarinic landscape ([LINK](#)), we went over the history of the muscarinic class in our note where we highlighted the origin and history of this class:

- The interest in muscarinic receptors from a clinical research perspective started with xanomeline which unexpectedly reduced psychotic symptoms in patients with AD (Bodick, N.C. et al., 1997), however, the initial development in this class had been limited due to tolerability issues that led to failure of the different development programs.
 - Xanomeline was assessed in both AD and schizophrenia and demonstrated improvement in both cognition and psychotic symptoms; however, the studies were discontinued following observation of GI side effects (~52% of patients discontinued treatment in the AD study) and peripheral cholinergic AEs, such as nausea, excessive sweating, and vomiting.
 - There were other drugs that also faced setbacks due to safety, examples of which include GSK1034702 (by GlaxoSmithKline) and AZD6088 (by AstraZeneca), both of which did not progress beyond Ph 1 due to safety and tolerability issues.
- Following the clinical observations with xanomeline, preclinical efforts began to understand the role of mAChRs (muscarinic acetylcholine receptors) in psychosis and cognition ([Figure 4](#)).
 - mAChR family consists of 5 subtypes (M1 - M5) with M1/M3/M5 receptor subtypes activating phospholipase C (PLC) whereas M2/M4 inhibit adenylate cyclase.
 - Within these receptor subtypes, the therapeutic development focus has mainly been on M1, which has early on emerged as a potential target in a range of brain disease based on M1-receptor KO mice model that demonstrated deficit in learning and memory (Bradley, S. J. et al. 2017; [Figure 3](#)).
 - In addition to M1, there is now growing body of evidence that has highlighted the therapeutic potential of M4-receptors (Foster, D. J., Choi et al., 2014). The M4-receptor KO mice model showed a hyper-locomotion phenotype and deficits in startle response, as well as elevated dopamine levels. In preclinical schizophrenia models, M4 receptor activation has also been shown to improve behavioral disturbances, including defective associated learning and memory (Bubser M. et al., 2014; Gould R.W. et al., 2018).

A big milestone for that class has been the approval of Cobenfy in schizophrenia in September 2024, which validated the class in the clinic and has paved the path for further development in the field.

Figure 4 - Timeline of key events in mAChR development



Source: Samantha E. Yohn et al., 2022

Abbreviations: AD, Alzheimer's disease; AP, antipsychotic; HPC, hippocampus; KO, knockout; NHP, nonhuman primate; PAM, positive allosteric modulator; PANSS, Positive and Negative Symptom Scale; PET, positron emission tomography; PFC, prefrontal cortex; PRZ, pirenzepine; SZ, schizophrenia.

Muscarinic agonists need to be combined with peripherally acting antagonists to avoid acetylcholine signaling outside the central nervous system

A muscarinic-based therapy needs to **activate muscarinic receptors in the brain to achieve therapeutic benefit**, i.e., increase **cholinergic signaling in the CNS (pro-cholinergic activity)**, while **avoiding excessive cholinergic stimulation in peripheral tissues**, which can produce intolerable side effects such as nausea, vomiting, sweating, diarrhea, and salivation. To mitigate these effects, peripherally restricted antimuscarinics such as Tropicium are combined with central muscarinic agonists to block peripheral receptors without entering the brain.

If muscarinic receptors are **overstimulated**, patients may experience **pro-cholinergic adverse events** such as **nausea, vomiting, diarrhea, and sweating**, which ultimately led to the discontinuation of xanomeline in clinic (Anantha Shekhar et al., 2008; Neil C. Bodick et al., 1997). These occur because of **acute peripheral muscarinic activation** and are typically **dose-dependent, transient, and reversible**, meaning clinicians can often **manage them with dose titration**.

Conversely, **excessive anticholinergic activity**—i.e., blocking acetylcholine signaling—can impair central cholinergic function and lead to **cognitive slowing, memory impairment, and worsening dementia**, particularly in vulnerable populations. A **high anticholinergic burden** has been associated with **falls, hospitalizations, increased mortality, and accelerated cognitive decline**.

Because acetylcholine is a key regulator of the **parasympathetic nervous system**, blocking muscarinic receptors can also produce **systemic anticholinergic effects**, including **constipation, urinary retention, dry mouth, blurred vision, tachycardia, and reduced sweating**, reflecting widespread parasympathetic inhibition.

This mechanism has been proven with the approval of Cobenfy in 2024, which demonstrated improvement over antipsychotics on both efficacy and safety

The MoA for ML-007 has been validated in clinic based on the regulatory studies and approval of Cobenfy

Cobenfy's MoA is a combination of xanomeline (M1 / M4 agonist, similar to ML-007) and trospium (non-selective antimuscarinic agent).

- The pivotal studies, EMERGENT-2 & -3, were initiated based on the positive Ph 2 EMERGENT-1 data where KarXT demonstrated a stat sig pbo-adjusted reduction of -11.5 on PANSS total score with an ES of 0.75 at week 5 ([Figure 5](#)). In addition, it demonstrated stat sig improvement on both PANSS positive and negative score. In comparison, the other therapies approved for schizophrenia have demonstrated improvement on PANSS total score ranging between -4.2 to -12.7 (the higher range is from an older study in 2002) with an ES of 0.3–0.5 ([Figure 6](#)).
- The approval of Cobenfy was based on the pivotal studies, EMERGENT-2 & -3, where it demonstrated a stat sig reduction of -9.6 and -8.4 on PANSS total score at week 5 with an ES of 0.61 and 0.6, respectively. On the SE, similar to Ph 2, it demonstrated stat sig improvement on PANSS Positive Score, although in the EMERGENT-3 study, it was not significant on the PANSS negative score.
- In terms of safety ([Figure 7](#)), there were some concerning AEs such as hypertension, (10% vs 1% in EMERGENT-2 and 6.4% vs 1.6% in EMERGENT-3) and GI related AEs such as vomiting (14% vs 1% in EMERGENT-2 and 16% vs 0.8% in EMERGENT-3), nausea (19% vs 6% in EMERGENT-2 and 19% vs ~2% in EMERGENT-3), diarrhea (6% vs 3% in EMERGENT-2 and 6% vs ~1% in EMERGENT-3), constipation (21% vs 10% in EMERGENT-2 and 12.8% vs ~4% in EMERGENT-3), and dyspepsia (19% vs 8% in EMERGENT-2 and ~6% vs 8% in EMERGENT-3). However, it was an improvement over the current SoC (SSRIs / SNRIs, antipsychotics) that can have weight gain, EPS, and sexual dysfunction ([Figure 8](#)).
- Additionally, none of safety concerns led to a warning or black box warning on the FDA label for Cobenfy, unlike the other approved therapies for schizophrenia.

Figure 5 - KarXT Efficacy Data

Therapies	Cobenfy (KarXT) xanomeline/trospium BMS/Karuna		
Studies	EMERGENT-1 (Ph2)	EMERGENT-2 (Ph3)	EMERGENT-3 (Ph3)
FDA Approval Year	2024		
Mechanism	M1/M4 muscarinic agonist + peripheral antagonist		
Approved Dose	50/20 mg BID to 125/30 mg BID		
Trial Duration (wks)	5	5	5
N (Drug / PBO)	90 / 92	125 / 128	125 / 131
Baseline PANSS (mean)	~97	98.3	~97
PANSS Total Score			
LS Mean Δ vs baseline (Drug)	-17.4	-21.2	-20.6
LS Mean Δ vs baseline (PBO)	-5.9	-11.6	-12.2
LS Mean Δ vs PBO	-11.6	-9.6	-8.4
p-value vs PBO	<0.001	<0.0001	<0.0001
Effect Size (Cohen's d)	0.75	0.61	0.60
PANSS Positive Subscale			
LS Mean Δ vs baseline (Drug)	-5.6	-6.8	-7.1
LS Mean Δ vs PBO	-3.2	-2.9	-3.5
p-value vs PBO	<0.001	<0.001	<0.001
PANSS Negative Subscale			
LS Mean Δ vs baseline (Drug)	-3.2	-3.4	NR
LS Mean Δ vs PBO	-2.3	-1.8	NS
p-value vs PBO	<0.001	0.0055	NS
PANSS Marder Neg Factor			
LS Mean Δ vs baseline (Drug)	-3.9	-4.2	-3.48
LS Mean Δ vs PBO	-2.6	-2.2	-0.82
p-value vs PBO	<0.001	0.0022	0.1957 (NS)

Source: FDA Label; Inder Kaul et al., 2024; Christoph U. Correll et al., 2022; Inder Kaul et al., 2024

Figure 6 - Efficacy Comparisons of KarXT vs other approved schizophrenia therapies

	Cobenfy (KarXT) xanomeline/trospium BMS/Karuna			Caplyta lumateperone Intra-Cellular	Vraylar cariprazine AbbVie	Rexulti brexpiprazole Otsuka/Lundbeck	Abilify aripiprazole Otsuka
	EMERGENT-1 (Ph2)	EMERGENT-2 (Ph3)	EMERGENT-3 (Ph3)	Study 005 / 301 (Ph2 / Ph3)	MD-16 / MD-04 / MD-05 (Ph2b / Ph3 / Ph3)	Vector / Beacon (Ph3 / Ph3)	Pivotal Trials (Ph3)
FDA Approval	2024			2019	2015	2015	2002
Mechanism	M1/M4 muscarinic agonist + peripheral antagonist			5-HT2A / D2 / SERT modulator	D3/D2 partial agonist 5-HT1A partial agonist	D2/5-HT1A partial agonist; 5-HT2A antag	D2 partial agonist 5-HT1A/5-HT2A
Approved Dose Range	50/20 to 125/30 BID			42 mg QD	1.5–6 mg QD	2–4 mg QD	10–30 mg QD
Trial Duration	5 weeks	5 weeks	5 weeks	4 weeks	6 weeks	6 weeks	4–6 weeks
N (Drug / PBO)	90 / 92	126 / 126	~128 / ~128	84 / 85 & 150 / 150	~140-155 / ~145-153 per arm	~180 / ~180 per study	~100-200 / ~100 per arm
Baseline PANSS (mean)	~97	98.3	~97	88.1 / 89.8	96.1–97.3	~95.6–95.8	~93–100
PANSS Total Score							
LS Mean Δ vs BL (Drug)	-17.4	-21.2	-20.6	-13.2 / -14.5	-16.0 to -22.3 (dose-dependent)	-16.6 to -20.7	-11.3 to -15.5
LS Mean Δ vs BL (PBO)	-5.9	-11.6	-12.2	-7.4 / -10.3	-9.0 to -14.3	-12 to -13.5	-2.3 to -5.3
LS Mean Δ vs PBO	-11.6	-9.6	-8.4	-5.8 / -4.2	-6.0 to -10.4 (dose-dependent)	-3.1 to -8.7	-5.9 to -12.7
p-value vs PBO	<0.0001	<0.0001	<0.0001	0.017 / 0.02	<0.001 (all doses 1.5–6mg)	<0.001	<0.001
Effect Size (Cohen's d)	0.75	0.61	0.60	0.4 / 0.3	~0.3–0.5 (dose-dependent)	~0.35–0.4	~0.3 to 0.5

Source: FDA Labels; Kaul et al. *Lancet* 2024; Kaul et al. *JAMA Psychiatry* 2024; Brannan et al. *NEJM* 2021; Correll et al. *JAMA Psychiatry* 2020; Durgam et al. *Schizophr Res* 2014 & *JCP* 2015; Kane et al. *JCP* 2015; ICER 2023

Figure 7 - KarXT Safety Data

Adverse Event (Drug vs PBO)	Cobenfy (KarXT) xanomeline/trospium BMS/Karuna		
	EMERGENT-1 (Ph2)	EMERGENT-2 (Ph3)	EMERGENT-3 (Ph3)
Nausea	17% / 4.4%	19% / 6%	19.2% / 1.6%
Vomiting	9% / 4.4%	14% / 1%	16% / 0.8%
Constipation	18% / 3.3%	21% / 10%	12.8% / 3.9%
Headache		14% / 12%	11.2% / 11.7%
Dizziness		9% / 3%	6.4% / 1.6%
Hypertension		10% / 1%	6.4% / 1.6%
Dyspepsia/GERD		19% / 8%	5.6% / 7.8%
Diarrhea		6% / 3%	5.6% / 0.8%
Dry Mouth	9% / 1.1%	19% / 8%	NR
Tachycardia		~5% / ~2%	~3% / ~1%
KEY DIFFERENTIATING AEs			
Weight Gain (≥7%)	2.2% / 5.6%	0% / 1%	NR
EPS / Akathisia		0% / 0% 1% / 1%	~1%
Somnolence/Sedation	7.9% / 6.7%	5% / 4%	NR
Prolactin Elevation		None observed	
Metabolic Effects (glucose/lipids)		None observed	
QTc Prolongation Risk		None	
Overall Discontinuation (Drug vs PBO)		25% / 21%	NR
D/C due to AEs (Drug vs PBO)		7% / 6%	6.4% / 5.5%

Source: FDA Label; Inder Kaul et al., 2024; Christoph U. Correll et al., 2022; Inder Kaul et al., 2024

Figure 8 - Safety Comparisons of KarXT vs other approved schizophrenia therapies

Adverse Event (Drug vs PBO)	Cobenfy EMERGENT-1	Cobenfy EMERGENT-2	Cobenfy EMERGENT-3	Caplyta 42mg	Vraylar 3-6mg	Rexulti 2-4mg	Abilify 10-30mg
Nausea	17% / 4.4%	19% / 6%	19.2% / 1.6%	9% / 5%	5 - 8% / 5%		15% / 11%
Vomiting	9% / 4.4%	14% / 1%	16% / 0.8%	3% / 2%	4 - 5% / 3%		11% / 6%
Constipation	18% / 3.3%	21% / 10%	12.8% / 3.9%	7% / 3%	6 - 10% / 5%		11% / 7%
Headache		14% / 12%	11.2% / 11.7%	19.3% / 15%	9 - 18% / 13%		27% / 23%
Dizziness		9% / 3%	6.4% / 1.6%	7% / 4%	3 - 5% / 2%		10% / 7%
Hypertension		10% / 1%	6.4% / 1.6%		up to 6% / 1%		<1%
Dyspepsia/GERD		19% / 8%	5.6% / 7.8%		4 - 5% / 4%	up to 6% / 2%	9% / 7%
Diarrhea		6% / 3%	5.6% / 0.8%		1 - 5% / 3%	3% / 2%	
Dry Mouth	9% / 1.1%	19% / 8%	NR	7.3% / 4.7%	1 - 3% / 2%		5% / 4%
Tachycardia		~5% / ~2%	~3% / ~1%		2 - 3% / 1%		2% / <1%
KEY DIFFERENTIATING AEs							
Weight Gain (≥7%)	2.2% / 5.6%	0% / 1%	NR	1% / 1.8%	2 - 3% / 1%	up to 4% / 2%	2% / 1%
EPS / Akathisia		0% / 0% 1% / 1%	~1%	4% / 2.7%	up to 34% / 12%	up to 7% / 5%	18% / 6%
Somnolence/Sedation	7.9% / 6.7%	5% / 4%	NR	17.3% / 11.3%	up to 10% / 5%	up to 3% / 1%	12% / 7%
Prolactin Elevation		None observed		Minimal	Minimal	Minimal	Minimal
Metabolic Effects (glucose/lipids)		None observed		Minimal	Mild	Mild	Minimal
QTc Prolongation Risk		None		Minimal	Low	Low	Low

Source: FDA Label; Inder Kaul et al., 2024; Christoph U. Correll et al., 2022; Inder Kaul et al., 2024

Muscarinic receptor agonists in addition to impacting positive symptoms, could improve negative symptoms and cognition, without the EPS and metabolic AEs associated with antipsychotics

The muscarinic hypothesis of schizophrenia emerged from observations that clozapine's efficacy may involve muscarinic receptor activity, that muscarinic agonists such as xanomeline reduced psychosis in Alzheimer's trials, and that post-mortem studies showed reduced muscarinic receptor signaling in schizophrenia.

Activation of M4 muscarinic receptors in the striatum indirectly suppresses dopamine signaling, which helps reduce positive symptoms (hallucinations, delusions), without the histamine H1 receptor and dopamine D2 receptor blockade, which helps avoid the EPS and metabolic side effects. Further, activation of M1 receptors can help enhance cholinergic signaling involved in cognition, thereby improving negative symptoms (apathy, social withdrawal) as well as cognitive deficits (working memory, attention).

The pivotal EMERGENT clinical trials (EMERGENT-1, -2, and -3) evaluated Cobenfy in patients with schizophrenia experiencing acute psychosis. Cobenfy showed statistically significant improvements in PANSS positive subscale scores, reflecting reductions in hallucinations, delusions, and other psychotic symptoms. It also demonstrated a statistically significant improvement on PANSS negative score in the Ph 2 EMERGENT-1 and Ph 3 EMERGENT-2 study, but the effect was not statistically significant in the Ph 3 EMERGENT-3 study.

While Cobenfy has been a meaningful improvement over SoC, it still leaves room for improvement on several fronts

- **High rates of both pro- and anticholinergic side effects:** This may have been caused by a mismatch of agonist and antagonist exposures in the periphery. To mitigate these cholinergic side effects, Cobenfy requires inconvenient dosing regimens (frequency, titration and fasting requirements) that are likely to result in patient compliance and adherence challenges. This was validated by our physician call who noted that in her patients, the dose titration on starting dose can last up to few weeks and yet to reach the maximum dose of 125 / 30 mg BID in any of her patients.
- **Inconvenient dosing:** Cobenfy has a BID dose in schizophrenia and is being assessed with a TID dose in AD psychosis. This can cause issues with compliance as the compliance rates go down with each additional dose (~79 – 91% for QD, ~69 – 73% for BID, and ~65% for TID / multiple daily dosing; Michael Iskedjian et al., 2002, AJ Claxton et al., 2001). Besides the dosing schedule, Cobenfy requires

fasting for at least an hour before a meal or at least two hours after a meal and dose titration during initiation which can last anywhere from 7 days to a few weeks.

- **Limited proven benefit on cognition:** Furthermore, although exploratory analyses in these trials suggested a positive effect on cognition symptoms in patients with baseline cognitive impairment, these analyses were not adequately powered to assess statistical significance.

INTELLECTUAL PROPERTY

ML-007C-MA has issued U.S. patents which are expected to expire between 2031 and 2032 along with a potential extension up to 2044–2045 based on additional pending applications

- For ML-007C-MA, there are 3 issued U.S. patents covering composition of matter and one issued U.S. patent covering use of ML-007 to treat, among other things, schizophrenia and ADP.
- The issued patents covering the composition of matter of ML-007 are expected to expire between 2031 and 2032, exclusive of possible patent term adjustments or extensions or other forms of exclusivity, and the issued patent covering the use of ML-007 to treat schizophrenia and ADP is expected to expire in 2031, exclusive of possible patent term adjustments or extensions or other forms of exclusivity.
- There is also one pending PCT (Patent Cooperation Treaty) directed to pharmaceutical compositions containing ML-007 and a PAC, and use of the composition to treat schizophrenia and ADP. If granted, this will be expected to expire in 2044, exclusive of possible patent term adjustments or extensions or other forms of exclusivity.
- Lastly, there is one pending U.S. provisional application covering salts and crystalline polymorphs of ML-007 which, if granted, will be expected to expire in 2045

KEY INVESTOR DEBATES

KEY DEBATE #1: HOW CAN ML-007 BE DIFFERENTIATED IN THE MUSCARINIC CLASS?

ML-007 can potentially emerge with a superior clinical profile than Cobenfy and other muscarinics in development based on preclinical data

ML-007 appears to be on the right side of the debate where both M1 and M4 receptors are considered to be important for driving efficacy.

Several preclinical studies have suggested that both M1R and M4R may be necessary for driving efficacy

- A paper published in 2009 ([LINK](#)) measured the impact on xanomeline in both M4 and M1 KO mice models compared to WT mice model. Here, xanomeline attenuated the amphetamine-induced activity in WT mice ([Figure 15](#) , [Figure 16](#)), however, this effect is abolished in M4 KO mice ([Figure 16](#)) and partially attenuated in case of M1 KO mice ([Figure 16](#)) **suggesting a larger role for M4R but a minor role for M1R as well to drive xanomeline's activity.**
- A recent 2025 Nature paper ([LINK](#)) for ML-007's (by Maplight Therapeutics) preclinical data suggested that ML-007 and xanomeline require both M1 and M4 receptors to drive efficacy based on the ED50 in WT mice which was different from the ED50 in M1 KO and M4 KO mice ([Figure 17](#)). This suggests that **agonism at both M₁ and M₄ subtypes contributes to efficacy in preclinical models** . Similar to the 2009 paper, it seems that for xanomeline, M4R seems to play a larger role compared to M1R (WT: 2.75 mg/kg versus ~1.9X and ~2.35X shift in ED50 concentration for M1 and M4 KO, respectively).

Clinical experience so far suggests that an M1R plus M4R strategy is most de-risked

The only approved muscarinic on the market is Cobenfy, which demonstrated strong consistent efficacy across its EMERGENT program. Conversely, while emraclidine (by Cerevel Therapeutics / Abbvie) **an M4 PAM, showed promise in Phase 1b, it eventually ended up failing to demonstrate efficacy in larger Ph 2 studies** ([LINK](#)). We also saw a positive Ph 2 study from direclidine (by Neurocrine

Biosciences) which is an orthosteric M4 receptor. Both emraclidine and direclidine are being evaluated in additional studies, and they could very well also prove successful.

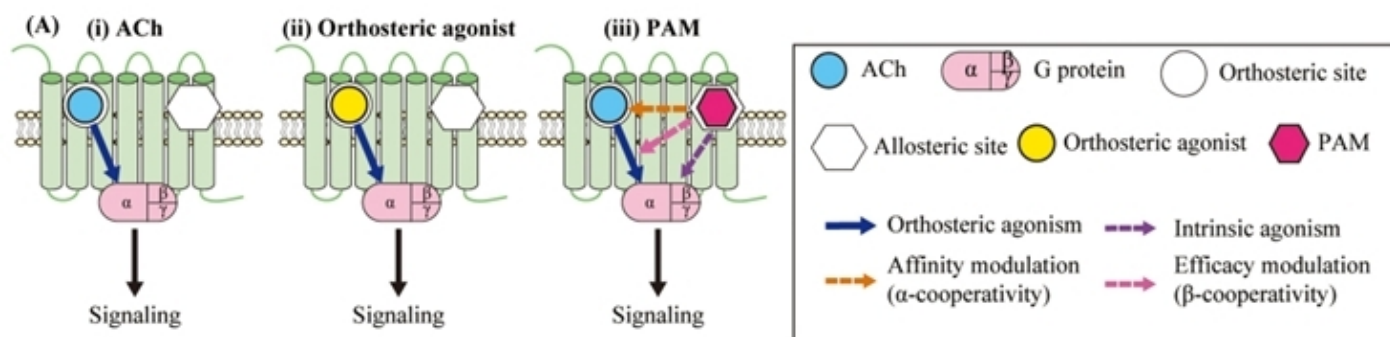
Festerodine is a better PAC than trospium due to its lack of food effect and potency.

Festerodine does not have food restrictions like trospium and is generally a more potent molecule (higher efficacy in two 12-week studies for overactive bladder) than trospium (Scott C. McFerren et al., 2015), however, it can result in higher AE % of dry mouth (up to 34.6% for festerodine vs up to 11.1% for trospium; FDA label) but this did not ultimately impact the discontinuation due to AE which is comparable between the two (~1% for both)

Two approaches to targeting the receptors are being explored, although each has its limitations – Orthosteric agonist (challenges with selectivity resulting in dose-limiting safety); PAM (limited by endogenous acetylcholine); so far, only orthosteric agonism (Cobenfy) has worked in clinic

This has previously been summarized in our note ([LINK](#)) where we highlighted the two main approaches being used by the different companies to target muscarinic receptors (LINK figure), namely orthosteric agonism and positive allosteric modulation (PAM). While orthosteric agonism has been traditionally used, PAM is a newer approach that is touted to be better as it can potentially avoid the safety issues seen with the orthosteric approach, although this has yet to be validated in clinic.

Figure 9 - Target Approach



Source: Takao Mandai et al., 2025

Orthosteric Agonist

An orthosteric agonist binds to the orthosteric site, the primary binding site on the muscarinic receptor, the same place where the natural ligand, acetylcholine binds, to activate the downstream signaling pathway. Essentially, the orthosteric agonist mimics ACh to activate the receptor.

A big challenge with developing a molecule that selectively targets the orthosteric binding site has been that the structure of the orthosteric binding site is well conserved across the mAChR subtypes (David M. Thal et al., 2016).

All five muscarinic receptor subtypes (M1, M2, M3, M4, M5) share a very similar shape, amino-acid composition, and 3D architecture at the orthosteric site, the pocket where the natural ligand acetylcholine (ACh) binds. Because this site is so structurally conserved, it looks nearly identical across all receptor subtypes. This functional conservation reflects evolutionary pressure to maintain ACh signaling across tissues.

This can result in binding of the drug to other receptors (such as M2/M3) which are highly expressed in peripheral tissues and result in dose-limiting cholinergic side effects related to the heart/GI (relating to M2/M3) such as GI cramping, diarrhea, salivation / sweating, hypotension, etc.

For xanomeline, the addition of trospium (LINK), now known as KarXT or Cobenfy, reduced these limitations. Trospium is a pan-muscarinic receptor antagonist that is restricted to the periphery and mitigates the cholinergic AEs that were otherwise resulting in treatment discontinuations.

The arguments in favor of using an orthosteric approach is that it has already had some success in the clinic, and does not require the presence of ACh

- One strong argument for using this approach is that this approach has already been proven in the clinic, as it was used for FDA approved Cobenfy, and is also being pursued for other therapies in development (highlighted in the sections below).
- Even though muscarinic receptors (M1–M5) are highly conserved, subtle residue changes in amino acids may be leveraged to drive selectivity, although the concern from a drug development point of view is that these differences may not be enough to drive strong selectivity.
- In-vitro data (Figure 14) for some orthosteric drugs in development highlights the low selectivity at other M subtypes, however, it is still possible that this results in some safety issues or dose-limiting issues in clinic (highlighted in the prior failures section).
- Since it directly activates the receptor, it does not require ACh to be present.

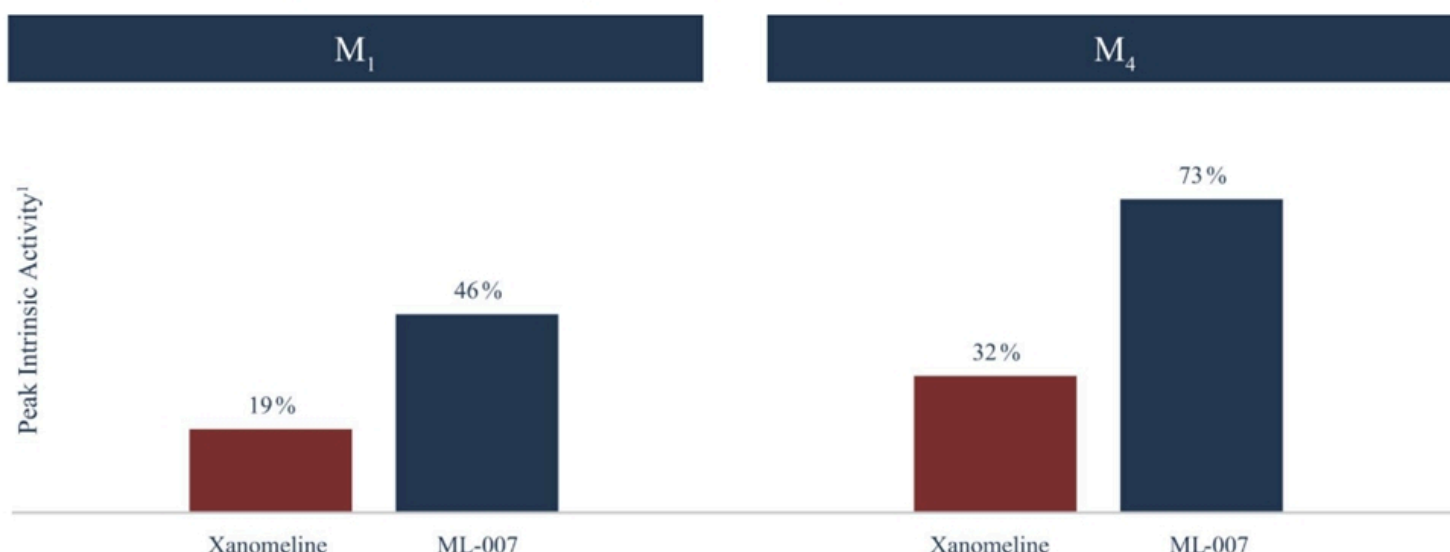
Positive Allosteric Modulator (PAM)

- To overcome this potential restriction, a positive allosteric modulator (PAM) approach can be used instead where the PAM binds to a different site, an allosteric site on the receptor which are less conserved among subtypes. It does not activate the receptor directly, instead, it enhances the effect of ACh by increasing its binding affinity to the receptor or increasing the receptor response to ACh.
- For instance, an M1R PAM binds to the allosteric site of the M1R and induces a conformational change in the M1R, resulting in an increase in the binding affinity between the natural ligand, acetylcholine, and the M1R and / or enhances the efficacy of acetylcholine (Takao Mandai et al., 2025).
- PAM has its own set of challenges which includes the requirement of endogenous acetylcholine that typically diminishes with age (GE Gibson et al., 1981). There is decrease in the synthesis and release of acetylcholine with age and this is more pronounced in neurodegenerative conditions like Alzheimer's disease.
- Additionally, even selective PAM can induce cholinergic side effects as both M1 and M4 are expressed in lower levels in the periphery (Takao Mandai et al., 2025)

Higher Relative Peak Intrinsic Activity at M1 and M4 Receptors for ML-007 vs Xanomeline

- In an in vitro GTPyS assay, ML-007 demonstrated strong partial agonist at both the M1 and M4 receptors, along with >2x peak intrinsic activity relative to xanomeline

Figure 10 - Relative Peak Intrinsic Activity at M1 and M4 Receptors for ML-007 and Xanomeline

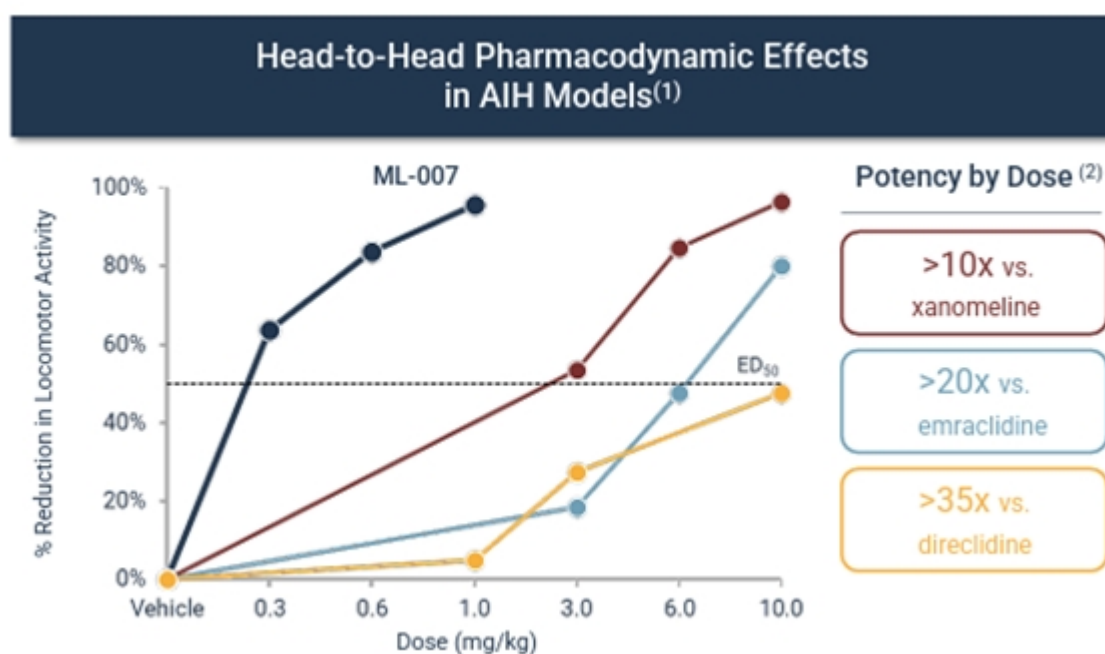


1. Represents data normalized to responses of control agonist, oxotremorine, in human GTPγS M₁ and M₄ in vitro assays.

Source: Company reports

- In the same preclinical studies as described above, xanomeline (active component of Cobenfy) was also assessed in the preclinical models of AIH, PCP-induced hyperlocomotion, and conditioned avoidance response where it demonstrated similar dose-dependent clinical activity, however, with a potency ten-fold higher than the dose of ML-007 required in all models. However, we do not know how this will translate in the clinic.
- Similarly, in AIH models, it has also demonstrated higher potency compared to emraclidine (>20X) and direclidine (>35X)

Figure 11 - ML-007 Potency Comparisons in AIH models

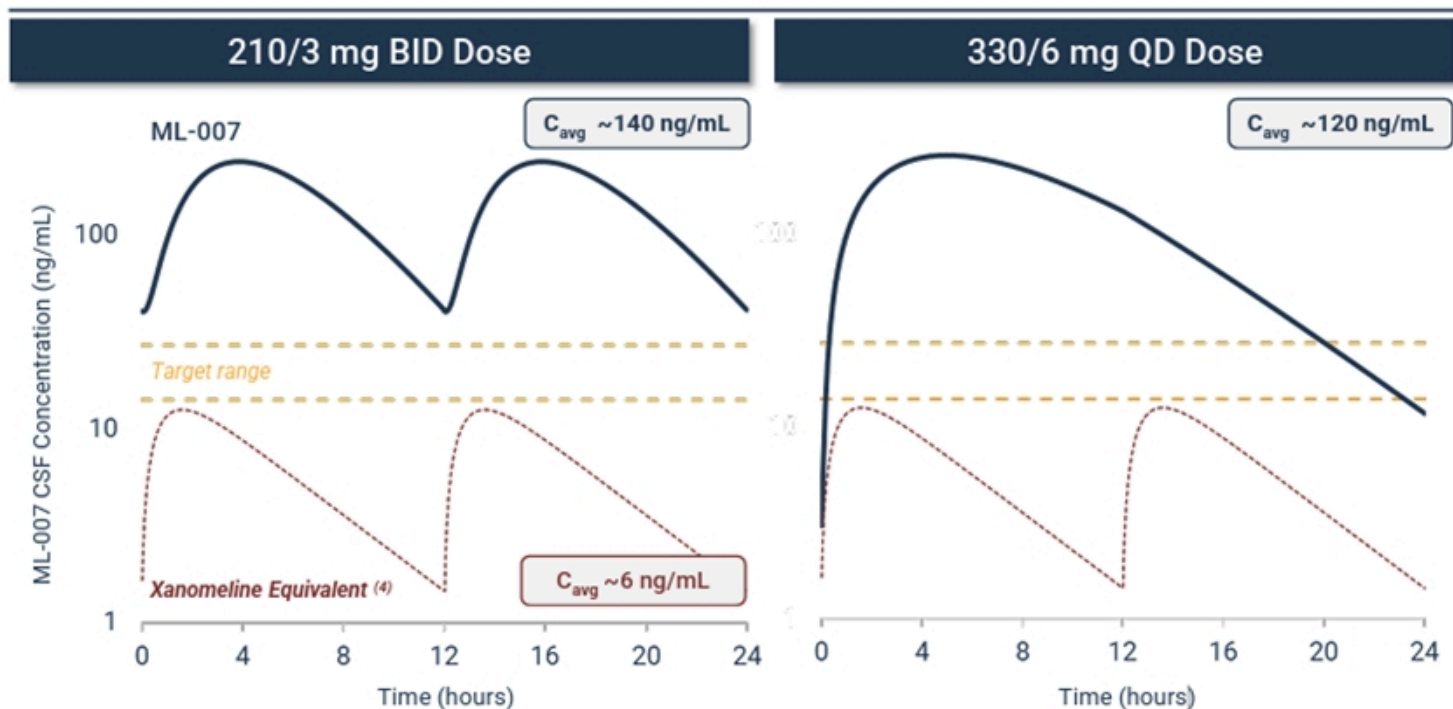


Source: Company Presentation

Higher steady state CSF exposure compared to xanomeline

- In addition to the higher potency in preclinical studies, ML-007, at the target Ph 2 doses, also demonstrated higher steady state CSF cons. compared to xanomeline
- Overall, this suggests that ML-007 is a more potent molecule compared to Cobenfy and some of the other muscarinics in development for schizophrenia (emraclidine and direcline)

Figure 12 - CSF Exposures for Ph 2 Target Doses



Source: Company Presentation

Lead asset, ML-007C-MA, has been strategically developed and formulated to achieve an optimal agonist/antagonist ratio through extensive preclinical and Ph 1 work

ML-007C-MA, is a fixed-dose combination of an M1/M4 muscarinic agonist, ML-007, co-formulated with a peripherally acting anticholinergic

Maplight acquired ML007 along with a portfolio of proprietary M1/M4 agonist molecules from NeuroSolis in June 2020 for an upfront payment of \$150K and is liable to issue NeuroSolis up to 1,043,014 shares, contingent upon development and regulatory milestones, of which 447,006 were issued in June 2025.

ML-007C-MA, is a fixed-dose combination of an M1/M4 muscarinic agonist, ML-007, **co-formulated with a peripherally acting anticholinergic (PAC)**, fesoterodine, which is being developed for the treatment of schizophrenia and Alzheimer's disease psychosis, or ADP.

ML-007C-MA is **designed to activate both M1 and M4 muscarinic receptors** in the CNS to drive efficacy, while synchronizing the pharmacokinetics of the agonist and antagonist components to mitigate peripheral cholinergic side effects.

ML-007 is highly soluble, is quickly absorbed and distributed throughout the body and has low protein binding which should support **low inter-patient variability** in clinical trials.

Festerodine, is an antimuscarinic medication with low CNS permeability used to treat overactive bladder symptoms, including urge continence, urgency, and frequency. This is similar to xanomeline which was coformulated with trospium, a pan-muscarinic receptor antagonist, and led to reduction in procholinergic AEs (63.6% with xanomeline vs 34.3% with xanomeline +trospium; Link table).

Fesoterodine was considered ideal for being used as a PAC based on its PK properties, low CNS permeability and minimal drug-drug interactions.

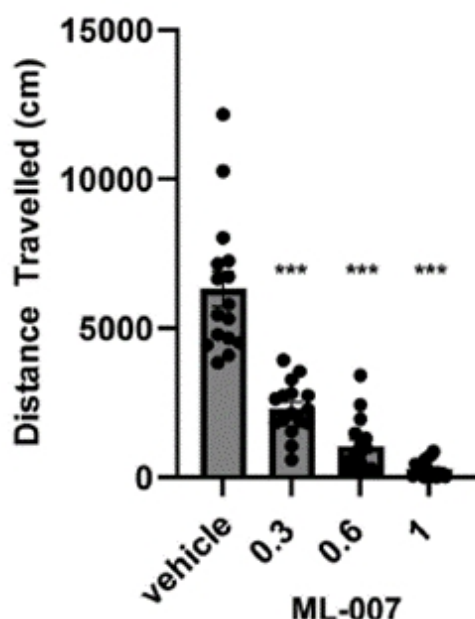
- Fesoterodine is highly soluble (> 50 mg/mL in water), has a **half-life of five hours and a Tmax of five to six hours with minimal drug-drug interaction liability**. This makes it ideal for a combination product, allowing for precision matching of the PK profiles.
- Fesoterodine, has demonstrated **low CNS permeability** in multiple third-party vitro and in vivo studies; in Maplight's dose-escalating animal studies, it demonstrated plasma ratios of ML-007 to PAC >80, suggesting no meaningful inhibition of ML-007's central effects.
- The PK of ML-007 and PAC were approximately **linear** and the **PK exposure was not affected by different dose combinations**. The steady-state ML-007:PAC concentration ratios after multiple repeated BID or QD dosing generally remained within the desired range of 100:1 to 600:1 over the majority of the dosing interval.

Preclinical data supports utility of ML-007C's mechanism of activating both M1 and M4 receptors in treating psychosis

ML-007C-MA is a brain-penetrant muscarinic agonist at both M1 and M4 muscarinic receptors that has demonstrated compelling efficacy across a range of preclinical models of psychosis in schizophrenia including amphetamine-induced hyperlocomotion, PCP-induced hyperlocomotion, and conditioned avoidance response (Susmita Chatterjee et al., 2025)

- Both M1 and M4 receptors have been implicated in CNS-related diseases with **M1 being related to learning and memory** (Bradley, S.J. et al. 2017) whereas **M4 receptor activation has been shown to improve behavioral disturbances** (Bubser M. et al., 2014).
- In AIH (amphetamine-induced hyperlocomotion) model, ML-007 exhibited a reduction of hyperlocomotion induced by amphetamine at all tested doses (0.3, 0.6, and 1 mg/kg). AIH is a widely used preclinical model that may mimic some aspects of psychosis (positive symptoms such as hallucinations and delusions) by inducing a transient hyperdopaminergic state ([Figure 13](#))

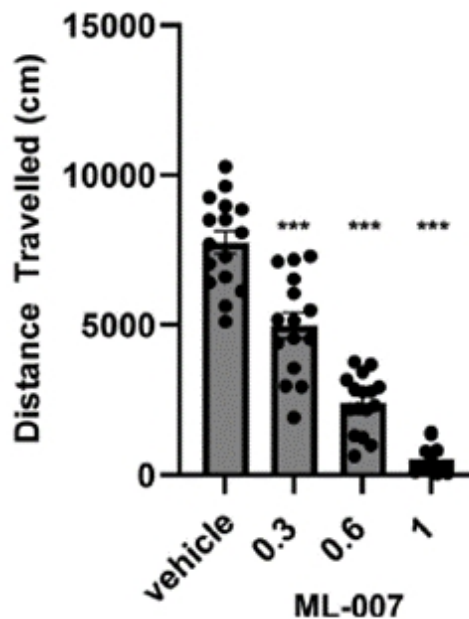
Figure 13 - Dose-dependent effects of ML-007 on reversal of AIH in mice



Source: Susmita Chatterjee et al., September 2025

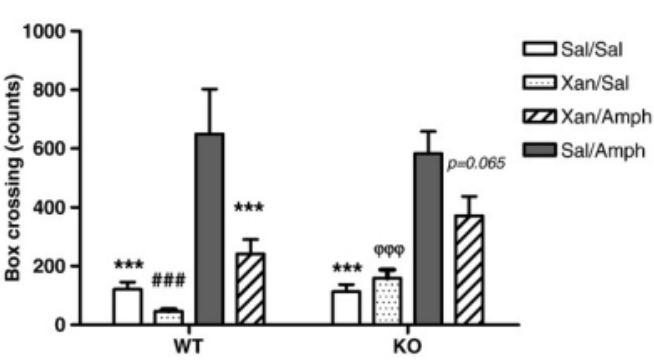
- Similarly, it demonstrated a dose-dependent reduction of PCP-induced hyperlocomotion in mice. PCP-induced hyperlocomotion is used in research as a model for positive symptoms ([Figure 14](#)). It involves the administration of phencyclidine (PCP), which causes a significant increase in motor activity and movement (hyperactivity) in animals.

Figure 14 - Dose-dependent effects of ML-007 on reversal of PCP-induced hyperlocomotion in mice



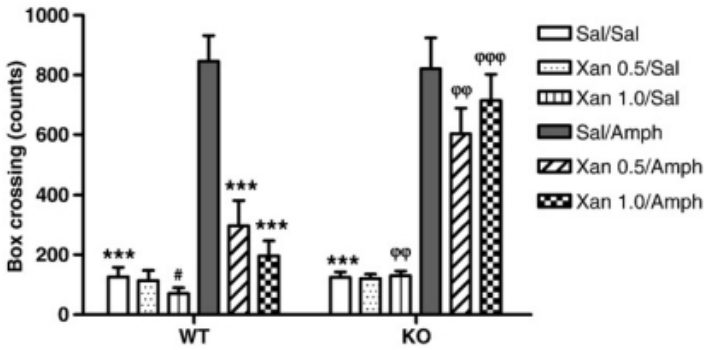
Source: Susmita Chatterjee et al., September 2025

Figure 15 - Xanomeline attenuated amphetamine-induced activity in WT mice, this effect partially attenuated in M1 KO mice



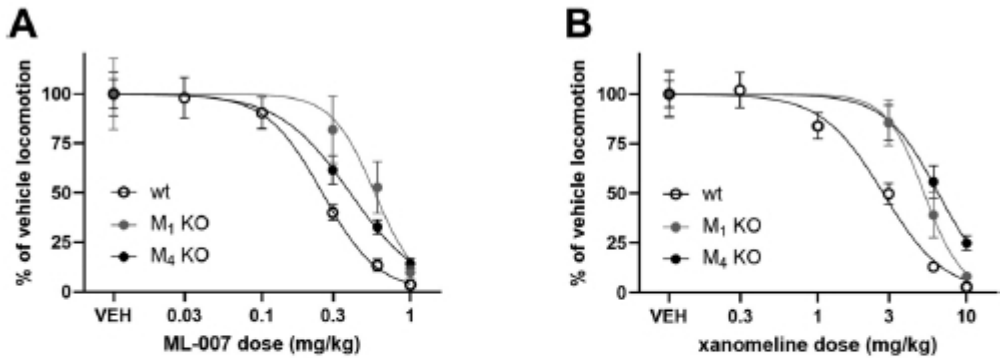
Source: Marie L. Woolley et al., 2008

Figure 16 - Xanomeline attenuated amphetamine-induced activity in WT mice, this effect was abolished in M4 KO mice



Source: Marie L. Woolley et al., 2009

Figure 17 - AIH Assay: ML-007 and Xanomeline appear to have activity on both M1 and M4; hwr, Xanomeline appears to have greater effect on M4, whereas ML-007 has greater effect on M1

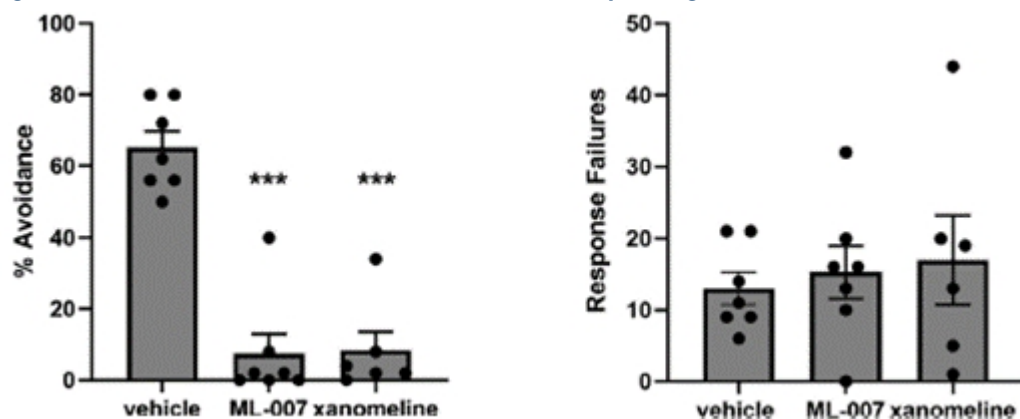


	Mouse strain	ED ₅₀ (mg/kg)	95% CI (mg/kg)
ML-007	WT	0.26	0.24 – 0.27
	M ₁ KO*	0.58	0.37 – 0.81
	M ₄ KO*	0.39	0.36 – 0.43
Xanomeline	WT	2.75	2.20 – 3.36
	M ₁ KO*	5.20	4.82 – 5.58
	M ₄ KO*†	6.47	6.04 – 6.93

Source: Susmita Chatterjee et al., 2025

- Lastly, on preclinical assays such as CAR assay (used to predict the antipsychotic potential of drug candidates), 1mg/kg of ML-007 demonstrated inhibition of avoidance response without impairing escape response behavior which has also been repoted by approved antipsychotic drugs.
- Overall, ML-007 has demonstrated compelling efficacy in animal models supporting further development in clinic.

Figure 18 - ML-007 attenuated conditioned avoidance responding at a tenfold lower dose than xanomeline

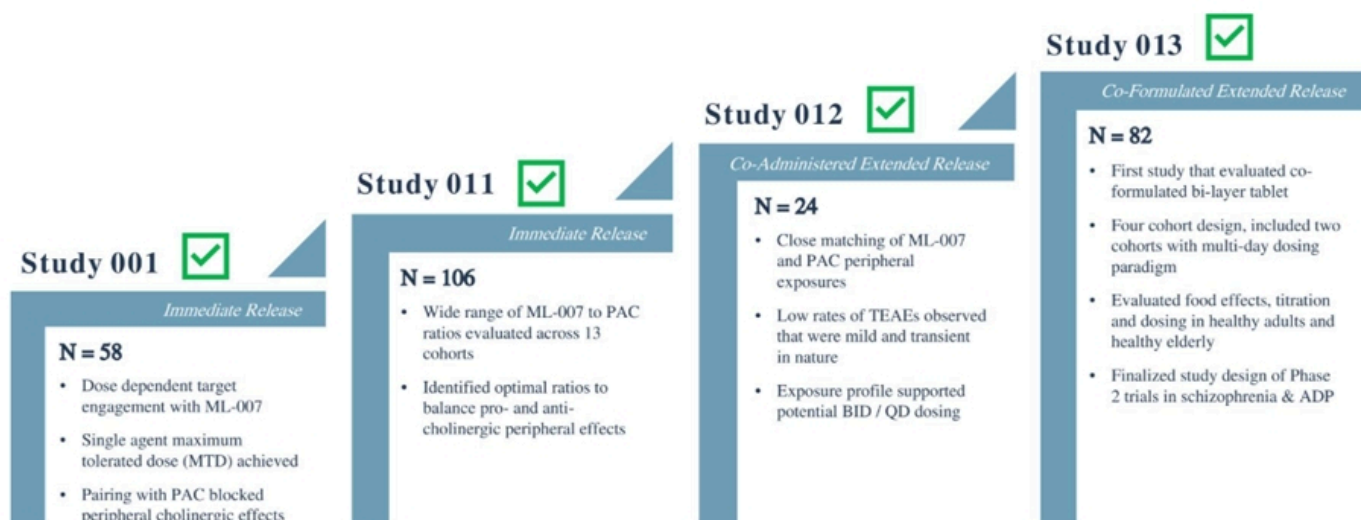


Source: Susmita Chatterjee et al., September 2025

Target Phase 2 doses for ML-007 + PAC were rationally selected based on several Phase 1 studies which helped identify a tolerable dosing regimen that provides adequate CNS exposure while managing peripheral cholinergic side effects.

ML-007 alone, co-administered, or co-formulated with PAC has been evaluated in four Phase 1 trials, with a total of 270 healthy participants enrolled and more than 1,500 doses of ML-007 administered to characterize the impact on safety and tolerability. These trials allowed for the establishment of well-tolerated dosing regimens expected to provide adequate CNS exposure of ML-007 while mitigating the peripheral cholinergic AEs.

Figure 19 - Ph 1 Studies



Source: S-1

ML-007C-MA Study 013

Figure 20 - Study 13 Cohorts

	Dose	Dosing	Subjects	N
Cohort 1	165/3mg BID	SAD under fast and fed	HV	10
Cohort 2	210/3 BID, 330/6 mg QD	Escalating doses	HV	8
Cohort 3	Multiple doses, 2 doses vs PBO	Multiple doses (165/3 mg BID, 210/3 mg BID, 270/6 mg QD, and 330/6 mg QD) with 7-day maintenance dose	HV	32, 8 in each dosing group
Cohort 4	Multiple doses, 2 doses vs PBO	Multiple maintenance doses of 165/3 mg BID, 210/3 mg BID and 330/6 mg QD	HV Elderly	32, 8 in each dosing group

Source: Company reports

- **Improved tolerability in fed conditions, no fasting needed.** ML-007 had higher PK in fed condition, while PAC had no food effect. Low vs high calorie had no impact on PK. ML-007C-MA was well tolerated in fed and fasted conditions, but tolerability was improved in fed conditions.
- **Improved tolerability of BID dose with a single titration dose.** Both 210/3 BID and 330/6mg QD doses were well tolerated; however, tolerability of the 210/3 mg dose improved when a single lower titration dose was administered 12 hours prior.
- **Most TEAEs were cholinergic in nature, and procholinergic events were reported more frequently than anticholinergic events.**
 - The most common non-procedural TEAEs (occurring in more than one participant in any ML-007C-MA group and at a greater incidence than placebo) during maintenance dosing were chills, constipation, dizziness, dyspepsia, headache, hyperhidrosis, nausea, salivary hypersecretion, and vomiting.
 - At the target doses selected for the ongoing Phase 2 ZEPHYR trial in schizophrenia, the most common non-procedural TEAEs (occurring in more than 1 participant in any ML-007C-MA group and at a greater incidence than placebo) during maintenance dosing are shown in Figure below. At these target doses, during the maintenance dosing period, no episodes of vomiting or constipation were reported.

Figure 21 - ML007-C-MA's Healthy Volunteer safety data for Ph 2 target doses

	Placebo Combined (N = 8) ⁽²⁾	210/3 mg BID (N = 6)	330/6 mg QD (N = 6) ⁽³⁾
Subjects With Any TEAE ⁽¹⁾	2 (25%)	3 (50%)	4 (67%)
Mild	2 (25%)	3 (50%)	4 (67%)
Moderate ⁽³⁾	0 (0%)	1 (17%)	2 (33%)
Severe	0	0	0
Most Common TEAEs (>1 Subject in any Dose Group)			
Chills	0	0	2 (33%)
Hyperhidrosis	0	1 (17%)	2 (33%)
Nausea	0	3 (50%)	3 (50%)
Dizziness	0	3 (50%)	2 (33%)
Dyspepsia	0	0	2 (33%)

Source: Company Presentation

■ **Phase 1 studies supported 210/3mg BID as the better tolerated dose in healthy elderly**

- In cohort 4, titration dose of 105/1.5 mg BID and maintenance doses of 165/3 mg BID and 210/3 mg BID were well tolerated. However, the maintenance dose of 330/6 mg QD was not well tolerated. Most TEAEs across the cohort were mild and there were no severe or serious TEAEs. Most TEAEs were cholinergic in nature, and procholinergic events were reported more frequently than anticholinergic events.
- In healthy elderly participants who received BID dosing regimens up to 210/3 mg, the most frequently reported non-procedural TEAEs (occurring in more than one participant in any ML-007C-MA BID group and at a greater incidence than placebo) during maintenance dosing were abnormal feces, hyperhidrosis, nausea, feeling hot, upper abdominal pain, dizziness, headache, penile burning sensation, and tremor. In elderly adults who received QD 330/6 mg, which was not considered well tolerated, the most frequently reported non-procedural TEAEs during maintenance dosing were nausea, dizziness, headache, tremor, hyperhidrosis, salivary hypersecretion, vomiting, chills, fatigue, malaise, increased blood pressure and decreased appetite.
- At the 210/3mg target dose selected for the ongoing Phase 2 trial in ADP, the most common non-procedural TEAEs (occurring in more than one participant in any ML-007C-MA group and at a greater incidence than placebo) during maintenance dosing are shown in Figure 10. At the target dose during the maintenance dosing period, only one episode of vomiting and one episode of constipation was reported. In addition, there were low rates of anticholinergic TEAEs, including no episodes of urinary retention.

Figure 22 - ML-007C-MA's Healthy Elderly Study - Safety Data for Ph 2 Target Dose

	Placebo Combined N = 7 ⁽²⁾	210/3 mg BID (with 2-7d titration) N = 11 ⁽²⁾
Subjects with Any TEAE ⁽¹⁾	5 (71%)	8 (73%)
Mild	5 (71%)	8 (73%)
Moderate	0	2 (18%)
Severe	0	0
Most Common TEAEs (>2 Subjects)		
Hyperhidrosis	0	3 (27%)
Nausea	0	3 (27%)
Dizziness	1 (14%)	3 (27%)
Headache	0	4 (36%)

Source: Company presentation

ML-007 is differentiated on safety compared to Cobenfy based on data in adults and elderly HV, especially on the AEs that result in tolerability issues such as vomiting and urinary retention (part of warnings on the FDA label).

Adults HV Study

To draw safety comparisons between Cobenfy and ML-007C-MA, we compare Cobenfy's Ph 1 MAD study in HV which evaluated five co-formulated doses of KarXT + trospium with Maplight's Study 013's cohort 3 in HV which evaluated four co-formulated ER doses of ML-007C-MA. Cobenfy has three approved doses in its label – 50/20 mg BID, an initiation dose for first two days, and two maintenance doses of 100/20 mg BID and 125/30 mg BID. The Ph 1 study included the initiation dose, the 100/20 mg BID dose, and a 125/40 mg BID dose, which is close enough to the approved dose, so we include this as well in our comparison. For ML-007C-MA, we focused on two of the four doses tested which are currently being studied in the Ph 2 schizophrenia study, the 210/3mg BID and the 330/6 mg QD.

The total TEAE rate (%) for Cobenfy appears to be slightly higher than that for ML-007, although the moderate AEs looked comparable.

- The TEAE rate for Cobenfy's maintenance doses, 110/20 BID and 125/40 BID, was 66.7% and 88.9%, respectively. In comparison, the TEAE rate ML-007's two doses being tested in Ph 2, 210/3 mg BID and 330/6 mg QD, was 50% and 66.7%, respectively.
- Technically the agonist to antagonist ratio for the highest Cobenfy dose tested (125 / 40 mg BID) in Ph 1 was lower than the approved dose (125 / 30 mg BID), i.e., 3.125 vs 4.166, which would mean, the peripheral cholinergic AEs for the approved dose are likely to be higher than what was seen in the Ph 1 study.
- The moderate AEs rates for the ML-007C-MA doses were 17–33% generally comparable to 5.7–33.3% rates seen for the Cobenfy maintenance doses. However, there could be some variability on the 125/40 mg BID dose of Cobenfy for reasons noted above.

It is hard to discern a trend in discontinuation rates from the HV studies due to limited information

- The treatment discontinuation rate for ML-007C-MA's dose 330 / 6 mg QD was higher at 17% (1/6 patients) than 6% (1/18 patients) for KarXT at 100/20 mg BID dose, but we have limited information on both. Maplight's management indicated to us that this discontinuation could not be directly attributed to any specific AE. The participant discontinued after mild AEs and one moderate AE

(dyspepsia), and while the investigator supported continuation, the participant withdrew consent. In the case of KarXT, the reason for the discontinuation was not disclosed.

- Outside the doses taken forward, we saw discontinuations in two of the KarXT doses, 1/18 in the 125/40 BID dose and 2/12 in the 150/40 BID dose. Again, the reason for discontinuation was not disclosed.

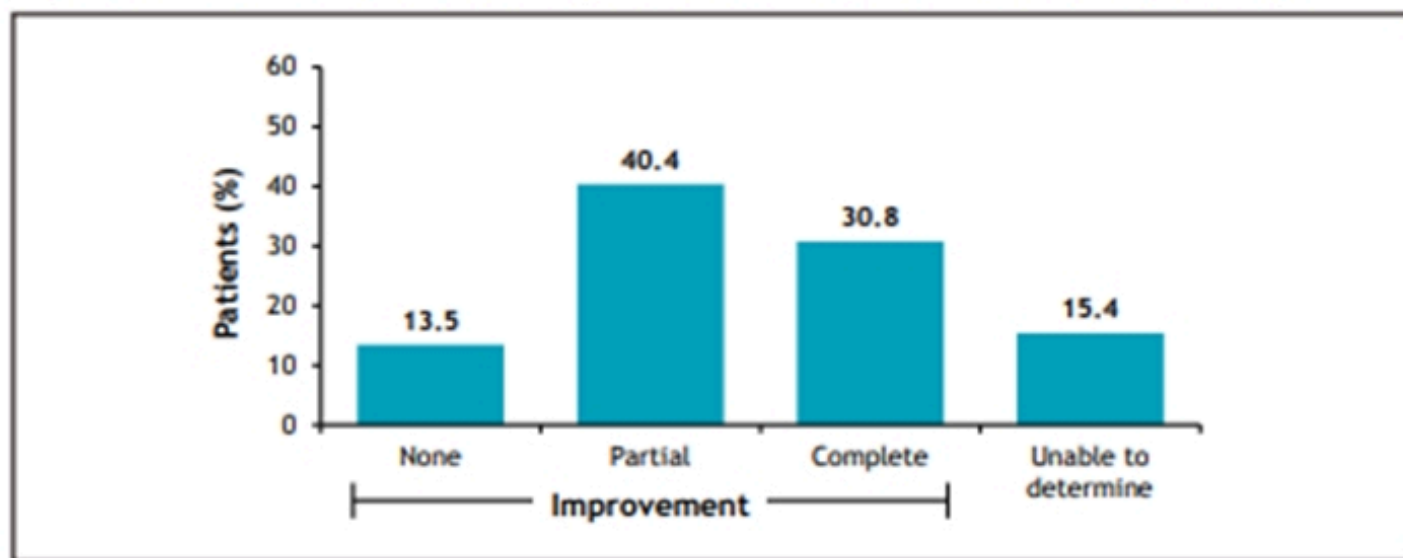
On procholinergic AEs, ML-007C-MA appears to be meaningfully differentiated on vomiting

The overall procholinergic AEs reported for ML-007C-MA were higher at 50% (210/3 mg BID) and 67% (330/6 mg QD) compared to Cobenfy's at 39% (110/20 BID) and 33% (125/40 BID); however, the main tolerability concern in the RW setting has been GI issues such as vomiting which was not seen with ML-007

- The nausea rate for ML-007C-MA (50% for both doses) appears to be worse than that for Cobenfy (22.2% for both doses), however, it is not clear how nausea was classified in the Cobenfy study as the rate of vomiting appears to be more than nausea in case of the 100 /20 mg BID cohort which seems counterintuitive to what has been seen with Cobenfy in the RW setting where the nausea rate is almost twice of vomiting (~65% of patients experience nausea and ~37% of patients experienced vomiting with Cobenfy; Christen Kutz et al., 2025).
- More importantly, the rate of vomiting for Cobenfy was 27.8% for the 100/20mg BID dose and 5.6% for the 125/40mg BID dose (this may be higher in a 125/30mg BID dose for reasons discussed earlier), whereas, there was no vomiting seen in the ML-007C-MA study.
- In addition, the RW data for Cobenfy presented at Psych Congress 2025 in a poster noted that amongst the patients who experienced nausea and vomiting, 55% of patients received pharmacological treatments and an antiemetic was used prophylactically or as needed in 21.1% and 44.4% of patients, respectively, with only ~30% of patients achieving complete improvement on these AEs.

Figure 23 - Effect of pharmacology tx on nausea and vomiting

Figure 5. Effect of pharmacologic treatment on nausea and vomiting



Source: Psych Congress Poster [September 2025]

ML-007C-MA appears to be meaningfully differentiated on anticholinergic AEs

- ML-007C-MA did not demonstrate any anticholinergic AEs in the HV study (Figure 24 at the target doses for Ph 2 compared to KarXT which reported AE rates of 16.7% (100/20 mg BID) and 66.7% (125/40 mg BID). The most common AEs reported included dry mouth, constipation, renal and urinary disorders, urinary retention and eye disorders. This could be a point of differentiation in the RW setting where patients prefer treatments with fewer side effects.

- An AEs of most concern here is **urinary retention** which has been listed under the Warnings and Precautions section of Cobenfy's label.
 - A total of 58 urinary retention adverse events (AEs) were reported among 1614 patients and volunteer subjects exposed to KarXT during product development, for an overall urinary retention AE incidence rate of 3.6% (58/1614).
 - The vast majority (95%) of the urinary retention events were reported in open-label, long-term studies (28/58 events; 48%) or completed phase 1 studies (27/58 events; 47%).
 - Most urinary retention AEs were mild (47/58 events; 81%) or moderate (7/58 events; 12%) in severity.
 - 4/57 events (7%) were reported as severe AEs. 2/57 events (3.5%) were reported as severe and serious. Of these, one patient required indwelling Foley catheter resulting in kidney injury due to KarXT-related urinary retention and post-renal azotemia. The second patient also required Foley catheter resulting in an incarcerated inguinal hernia and partial small bowel obstruction.
- While the rate of the severe AEs was not very high, this AE poses the risk of a patient being admitted to an ER for an elongated portion of treatment. The FDA review document for Cobenfy noted that KarXT-related urinary retention is a serious, potentially life-threatening risk and recommended it to be included as part of the warning and precaution section of the label ([LINK](#)).
- In case of elderly patients, there is limited data available, but some initial results from the Ph 1b study in healthy elderly patients suggest there was one serious AE of urinary retention in case of KarXT ([LINK](#)) whereas there are no episodes of urinary retention in case of ML-007C-MA group in healthy elderly cohort.
- ML-007 reported higher rates of hyperhidrosis at 17% (210/3 mg BID) and 33% (330/6 mg QD) than KarXT which reported AE rates of 5.6% (100/20 mg BID) and 11.1% (125/40 mg BID); however, we believe hyperhidrosis is less concerning of an AE, as it simply involves excessive sweating, which is not a tolerability concern for patients.

Overall, ML-007 appears to be more tolerable than KarXT especially given the issues with vomiting AEs in the RW setting as well as concerns around anticholinergic AEs such as urinary disorders that can be quite debilitating. Some questions that remain on safety are whether we will see higher rates of nausea and vomiting in a larger cohort as there was one patient who experienced vomiting in Cohort 4 (healthy elderly patients) and whether there will be any concerns around any anticholinergic AEs such as urinary retention in larger studies as the time to onset for this AE varies from several days to several weeks.

Figure 24 - ML-007C-MA vs KarXT: Healthy Volunteer Safety Data Comparisons

Therapy Company	ML-007C-MA Maplight Therapeutics			Cobenfy (KarXT) BMS				
Phase	Study 013			Ph 1 MAD				
Patient Population	Healthy Adults (Cohort 3)			Healthy Volunteers				
Study Design	1 - 4 dose titration + 7 days of target dose			The study design was comprised of a 2-day titration period of either placebo or a KarXT dose of 50 mg xanomeline + 20 mg tropism followed by a 5-day treatment period; The doses (all BID) assessed were: xanomeline 100 mg, 125 mg and 150 mg in combination with tropism 20 mg or 40 mg.				
Dose	Placebo	210 / 3 mg BID	330 / 6 mg QD	KarXT 50 /20 BID	Kar XT 100/20 BID	KarXT 150/20 BID	KarXT 150/40 BID	KarXT 125 /40 BID
Comments on doses	Ph 2 target dose			Approved dose	Approved dose			125/30 is the approved dose
N	8	6	6	53	18	5	12	18
SAFETY								
Any TEAE	25%	50%	67%	41.5%	66.7%	80.0%	83.3%	88.9%
Mild	25%	50%	67%	35.8%	50.0%	0.0%	33.3%	55.6%
Moderate	0%	17%	33%	5.7%	16.7%	80.0%	50.0%	33.3%
Severe	0%	0%	0%	0%	0%	0%	0%	0%
clinically significant TEAEs				5.70%	16.70%	80%	50%	33.30%
Drug-related TEAEs				39.60%	61.10%	80%	83.30%	88.90%
Mild				34%	44.40%		33.30%	55.60%
Moderate				5.70%	16.70%	80%	50%	33.30%
Severe				0%	0%	0%	0%	0%
Serious AE								
Study drug-related AE leading to discontinuation					5.6%		16.7%	5.6%
Discontinuations		One 330/6 mg participant discontinued; after mild AEs and one moderate AE (dyspepsia), the investigator						
Procholinergic	0%	50%	67%	11%	39%	40%	50%	33%
Gastrointestinal disorders				11%	39%	40%	50%	22%
Sweating				1.9%	5.60%	20%	0.0%	11.1%
Hyperhidrosis	0%	17%	33%	1.9%	5.6%	20%	16.7%	0%
Salivary hypersecretion				9.4%	22.2%	40%	41.7%	22.2%
Nausea	0%	50%	50%	0.0%	0%	0%	16.7%	0.0%
Diarrhea				1.9%	27.8%	20%	41.7%	5.6%
Vomiting		0%	0%					
Skin and subcutaneous tissue disorders				1.9%	5.60%	20%	0.0%	11.1%
Anticholinergic	13%	0%	0%	18.9%	16.7%	0.0%	58.3%	66.7%
GI disorders				18.9%	16.7%	0.0%	41.7%	27.8%
Dry mouth				13.2%	11.1%	0%	41.7%	27.8%
Constipation				5.7%	5.6%	0%	0.0%	5.6%
Dyspepsia				0%	0%	0%	16.7%	27.8%
Renal and urinary disorders				0%	0%	0%	0.0%	22.2%
Dysuria				0%	0%	0%	16.7%	11.1%
Urinary retention				0%	0%	0%	8.3%	27.8%
Eye disorders								
Others	13%	50%	50%					
Additional key AEs								
Syncope				7.5%	11.1%	0.0%	8.3%	11.1%
Postural Dizziness								
Most Common TEAEs								
Headache				3.8%	22.2%	80.0%	66.7%	38.9%
Tremor								
Chills	0%	0%	33%					
Hyperhidrosis								
Nausea								
Dizziness	0%	50%	33%	7.5%	11.1%	0.0%	8.3%	11.1%
Dyspepsia	0%	0%	33%					

Source: Company reports

Another key differentiation for ML-007 is on dosing convenience based on QD dosing option, minimal titration, and no fasting requirements

- The next key differentiation for ML-007 from Cobenfy is on dosing convenience where ML-007 **does not have any fasting requirements** like Cobenfy which is required to be taken at least 1 hour before or 2 hours after a meal
- Maplight is **likely to have more convenient doses** than Bristol in terms of dosing frequency, as it is evaluating both a QD and BID dose in its Ph 2 schizophrenia study, which could end up being at least as good if not better than Cobenfy's BID dosing. In its Ph 2 ADP study, it is evaluating a BID dose, which would be less frequent than the TID dose being studied for Cobenfy in ADP.
- Even on **dose titration**, Cobenfy's label requires a 3 to 8 day titration period with treatment being initiated at the lower dose of 50 /20 mg BID, which is a few weeks in the real world setting. On the contrary, ML-007 was well-tolerated in study 013 with minimal titration which was the first 12 hours at a lower dose followed by a higher maintenance dose in schizophrenia. In its Ph 2 ADP study, management decided to conservatively test a one-week long titration, although management noted that there is no difference between a shorter versus longer titration with ML-007C-MA.

There are limited comparative data points with the other muscarinic therapies in development, however, preclinical studies suggest higher potency for ML-007 compared to emraclidine and direclidine with a similar improvement on dosing schedule as ML-007C-MA has over Cobenfy and potentially comparable safety

There are other muscarinic therapies in development, including emraclidine by Abbvie, direclidine by Neurocrine Biosciences, and a muscarinic program by Neumora Therapeutics.

- **On efficacy**, we do not have clinical data in schizophrenia for ML-007C-MA, but preclinical comparisons demonstrated that ML-007 has higher potency than (>20X) and direclidine (>35X).
 - **Emraclidine had higher efficacy than KarXT in a small Ph 1b study, however, it failed in larger Ph 2 studies evaluating a 30 mg; higher doses are being tested by management.** Emraclidine is a selective M4 PAM that demonstrated compelling efficacy in Ph 1b study, however, it ultimately did not meet the PE in larger Ph 2 studies due to the larger than expected placebo response. Abbvie management is still looking at emraclidine in schizophrenia and is conducting MAD study to understand the dosing dynamics. Previous commentary has noted that its study has surpassed the 75 mg and is starting to test a 100 mg. The initial Ph 2 study had assessed 30 mg and these higher doses are being assessed to see the impact on receptor occupancy /clinical profile.
 - **Direclidine's appears to be comparable to KarXT based on ES but lower numerically in on PANNS change a smaller Ph 2 study, and so may ultimately end up having slightly lower efficacy than KarXT in larger Ph 3 studies ([Figure 25](#)).** Neurocrine's direclidine is an M4 agonist that reported positive Ph 2 data in August 2024 ([LINK](#)) and then additional data in May 2025 ([LINK](#)) where the data demonstrated comparable ES to KarXT (0.72 for '568 vs 0.75 for KarXT in Ph 2), although it showed a numerically lower the pbo-adjusted PANSS change at week 5 than KarXT (-10 for '568 vs -11.5 for KarXT). In terms of next steps, it is progressing in Ph 3 in schizophrenia and Ph 2 in bipolar mania.
 - **Neumora recently announced that it has selected NMRA-898** as the lead candidate for schizophrenia based on encouraging initial Ph 1 data, including human half-life of ~80–100 hours, supporting QD dosing. The company noted that it saw dose-proportional exposures with low variability and predicted free exposures in the brain which exceeded in-vitro M4 EC50 concentration. In terms of safety, there has been no blood-pressure elevation so far, although there were on-target heart rate increases similar to Cobenfy which confirmed target engagement.

Figure 25 - KarXT vs Direclidine: Efficacy Comparisons

	Cobenfy (KarXT) xanomeline / trospium chloride Bristol Myers Squibb			Direclidine (NBI-1117568) M4 selective agonist Neurocrine Biosciences	
	EMERGENT-1 (Phase 2)	EMERGENT-2 (Phase 3)	EMERGENT-3 (Phase 3)	SCZ2028 (Phase 2) 20mg QD	SCZ2028 (Phase 2) 20mg QD
DRUG CHARACTERISTICS					
Regulatory Status	FDA Approved (Sep 2024)			Phase 3	
Mechanism	Dual M1/M4 muscarinic agonist (xanomeline) + peripheral antagonist (trospium)			Selective M4 orthosteric agonist (no peripheral antagonist needed)	
Dosing Regimen	50/20 mg BID → 125/30 mg BID (flexible)			20 mg QD	20 mg QD
Trial Duration	5 weeks	5 weeks	5 weeks	6 weeks	5 weeks
N (Drug / PBO)	90 / 92	126 / 126	~128 / ~128	35 / 68	35 / 68
Total Enrolled	182	252	256	210 total (all arms + placebo)	
Age Range	18–60	18–65	18–65	18–55	
Baseline PANSS (mean)	~97	98.3	~97	NR (acute exacerbation; PANSS ≥80 likely)	
PANSS TOTAL SCORE					
LS Mean Δ from BL (Drug)	-17.4	-21.2	-20.6	-18.2	-20.2
LS Mean Δ from BL (PBO)	-5.9	-11.6	-12.2	-10.7†	-10.2
LS Mean Δ vs PBO	-11.6	-9.6	-8.4	-7.5	-10
p-value vs PBO	<0.0001	<0.0001	<0.0001	0.011	<0.001
Effect Size (Cohen's d)	0.75	0.61	0.60	0.61	0.72

Source: Brannan et al. NEJM 2021; Kaul et al. Lancet 2024; Kaul et al. JAMA Psychiatry 2024; Neurocrine Biosciences Press Releases

- On safety**, neither emraclidine nor direclidine have publicly reported any healthy volunteer data. We wait to see the clinical data for ML-007C-MA in schizophrenia to understand how the safety profile compares with these therapies. Additionally, emraclidine's safety profile could likely worsen now with the higher doses.
 - Direclidine's safety profile appears to be better than KarXT based on the Ph 2 data where there were no notable AEs of vomiting, diarrhea, or hypertension listed in the most common AEs (Figure 26)
 - The other AEs were lower for direclidine compared to KarXT such as constipation (5% for direclidine vs 18%, 21%, and 12.8% for KarXT) and nausea (5% for direclidine vs 17%, 19%, and 19.2% for KarXT), although some were higher such as dizziness (12.5% for direclidine vs 9% for KarXT) and somnolence (12.5% for direclidine vs 7.9% and 5% for KarXT)
 - Overall, direclidine's safety data appears to be an improvement from KarXT especially on the GI concerns and could potentially be a competitor for ML-007C-MA in terms of safety differentiation

Figure 26 - Direclidine's Safety Profile

	20mg QD N=40	40mg QD N=39	60mg QD N=34	30mg BID N=27	All Treated N=140	Placebo N=70
Somnolence	5 (12.5)	2 (5.1)	7 (20.6)	1 (3.7)	15 (10.7)	2 (2.9)
Dizziness	5 (12.5)	3 (7.7)	4 (11.8)	1 (3.7)	13 (9.3)	1 (1.4)
Headache	1 (2.5)	5 (12.8)	1 (2.9)	5 (18.5)	12 (8.6)	14 (20.0)
Nausea	2 (5.0)	3 (7.7)	3 (8.8)	0	8 (5.7)	2 (2.9)
Constipation	2 (5.0)	3 (7.7)	1 (2.9)	1 (3.7)	7 (5.0)	2 (2.9)

5.0% Treatment Discontinuation Rate Due to Adverse Events Across all Direclidine Arms vs. 4.3% For Placebo

Source: Company Presentation

- ML-007C-MA will have dosing convenience over Cobenfy; however, relative to other muscarinic therapies in development, its dosing convenience may not ultimately prove differentiating, as both emraclidine and direclidine are currently positioned as once-daily agents without titration requirements and with no food restrictions. However, with higher doses of emraclidine now under evaluation, the final dosing regimen remains to be determined.

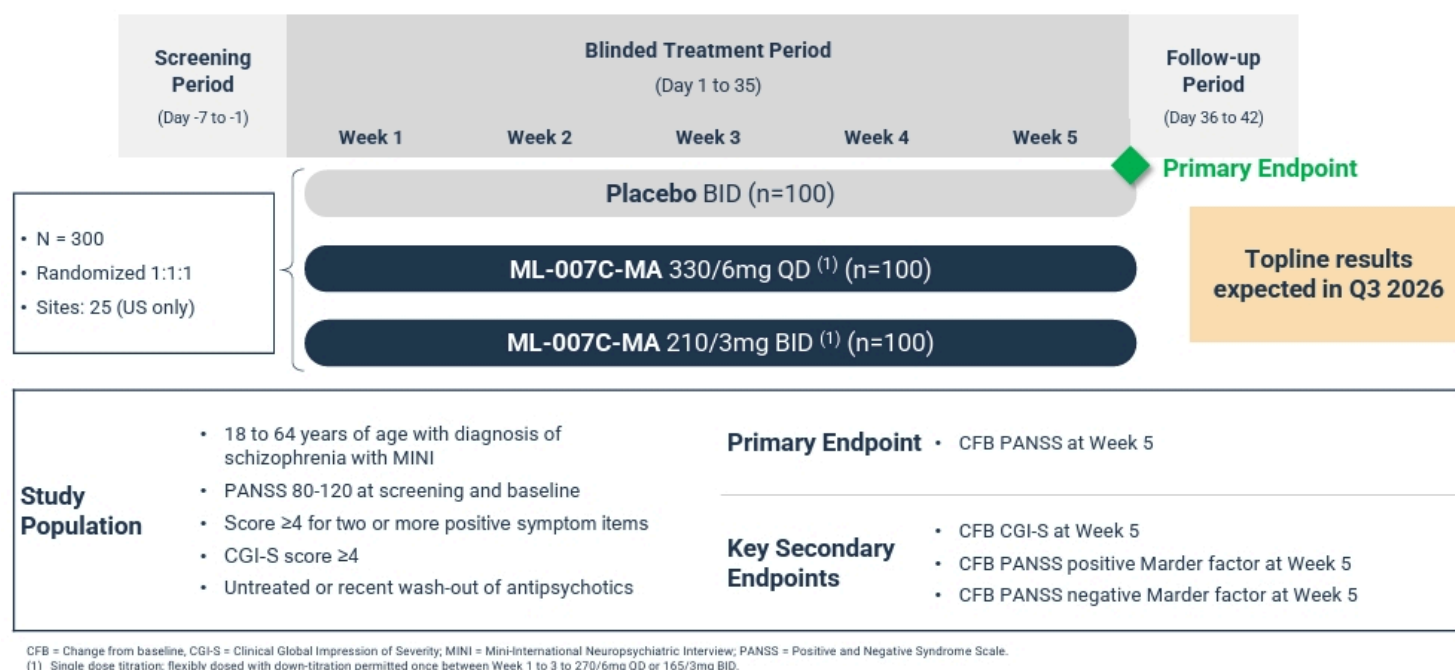
Overall, ML-007C-MA's differentiation from direclidine is narrower on dosing convenience and safety, but may have chances to be better on efficacy based on preclinical potency. For others, we are yet to see updated data from emraclidine's new doses and Neurmora's muscarinic program

In the upcoming Ph 2 readout, ML-007 can potentially demonstrate a better safety / tolerability profile than Cobenfy with a comparable ES to its Ph 3 studies

The main de-risking event this year for MPLT is the Ph 2 data readout for ML-007 in schizophrenia expected in 3Q26 (Figure 27).

- Similar to the previous muscarinic studies in schizophrenia, the PE is CFB (change from baseline) on PANSS score at Week 5.
- The sites enrolled in the study are US only which usually tend to have higher placebo responses in case of schizophrenia trials (Mathangi Gopalakrishnan et al., 2020); however, management selected US only sites because in its experience US sites are better form a data quality perspective and ex-US geography do end up having higher placebo responses barring Ukraine.

Figure 27 - ZEPHYR Ph 2 study design



Source: Company presentation

In terms of efficacy, the Ph 2 study has been 90% powered to demonstrate an ES comparable to KarXT's Ph 3 studies (ES of 0.61 and 0.6)

- The Ph 2 EMERGENT-1 had a higher ES of 0.75, however, that study likely benefitted from no expectation bias as it was the first study with a muscarinic agonist in patients.
- While we acknowledge that direclidine did demonstrate an ES of 0.72 at week 5 in the Ph 2 study, direclidine's Ph 2 study was much smaller with ~140 patients across 4 treatment arms (~34 patients at 60 mg QD dose) and 15 US sites ([NCT05545111](#)), whereas ML-007's Ph 2 study is a much larger study with 300 patients across 25 US sites, which in terms of size and geographic scale is more similar to the EMERGENT trials, which supports an ES of 0.6 being the more appropriate bar.

ML-007 should continue to demonstrate a better safety profile, particularly on vomiting and the overall anticholinergic AE rates as it demonstrated in the HV study, plus, we expect safety & tolerability to improve in schizophrenia patients, as seen with Cobenfy.

- During our conversation with management, it noted that the safety and tolerability profile could potentially improve in schizophrenia patients compared to healthy volunteers especially on a placebo-adjusted basis
- This was also seen with KarXT ([Figure 28](#)) where the AE% reduced from 88.9% at the highest test (125 / 40 mg BID) in Ph 1 HV study to 53.5%, 75%, and 70.40% at 125 / 30 mg BID dose in EMERGENT-1, -2, and -3, respectively.
- Even on the notable procholinergic and anticholinergic AEs, the rate of AEs were lower in schizophrenia patients barring constipation which was higher.
- We do note that there were higher rates of hypertension in the treatment arm vs placebo arm in the two Ph 3 studies which we will be looking out for in studies with ML-007.
- However, overall, we can potentially anticipate an improvement in the safety profile, however, given that there were anticholinergic AEs seen in the other dose cohorts of ML-007C-MA's Ph 1 study, we do not discount that there may be some anticholinergic AEs.

Figure 28 - KarXT (Cobenfy) Safety Data Across Studies

Therapy Phase Dose	Ph 1 MAD					KarXT (Cobenfy) Ph 2 EMERGENT-1		Ph 3 EMERGENT-2		Ph 3 EMERGENT-3	
	KarXT 50 /20 BID	Kar XT 100/20 BID	KarXT 150/20 BID	KarXT 150/40 BID	KarXT 125 /40 BID	KarXT	Placebo	KarXT	Placebo	KarXT	Placebo
Comments on doses	Approved dose	Approved dose			125/30 is the approved dose	50mg/20 mg BID on days 1-2, followed by 100 mg/20 mg BID on days 3-7, and increased to 125 mg/30 mg BID on days 8 - 34					
N	53	18	5	12	18	90	92	125	128	125	128
Discontinuations		5.6%		16.7%	5.6%			7%	6%	6.4%	5.5%
SAFETY											
Any TEAE	41.5%	66.7%	80.0%	83.3%	88.9%	53.50%	52.20%	75.0%	58.0%	70.40%	50.00%
Mild	35.8%	50.0%	0.0%	33.3%	55.6%	51.20%	41.30%				
Moderate	5.7%	16.7%	80.0%	50.0%	33.3%	14%	21.70%				
Severe	0%	0%	0%	0%	0%	2.30%	2.20%				
clinically significant TEAEs	5.70%	16.70%	80%	50%	33.30%						
Drug-related TEAEs	39.60%	61.10%	80%	83.30%	88.90%						
Mild	34%	44.40%		33.30%	55.60%						
Moderate	5.70%	16.70%	80%	50%	33.30%						
Severe	0%	0%	0%	0%	0%						
Serious AE								2.0%	2.0%	0.8%	0.0%
Procholinergic											
Gastrointestinal disorders	11%	39%	40%	50%	33%						
Sweating											
Hyperhidrosis	1.9%	5.60%	20%	0.0%	11.1%						
Salivary hypersecretion	1.9%	5.6%	20%	16.7%	0%						
Nausea	9.4%	22.2%	40%	41.7%	22.2%	17%	4%	19.0%	6.0%	19.2%	1.6%
Diarrhea	0.0%	0%	0%	16.7%	0.0%			6.0%	3.0%	5.6%	0.8%
Vomiting	1.9%	27.8%	20%	41.7%	5.6%	9%	4%	14%	1.0%	16%	0.8%
Skin and subcutaneous tissue disorders	1.9%	5.60%	20%	0.0%	11.1%						
Anticholinergic											
GI disorders	18.9%	16.7%	0.0%	41.7%	27.8%						
Dry mouth	13.2%	11.1%	0%	41.7%	27.8%	9%	1%				
Constipation	5.7%	5.6%	0%	0.0%	5.6%	18%	3%	21.0%	10.0%	12.8%	3.9%
Renal and urinary disorders	0%	0%	0%	16.7%	27.8%						
Dysuria	0%	0%	0%	0.0%	22.2%						
Urinary retention	0%	0%	0%	16.7%	11.1%						
Eye disorders	0%	0%	0%	8.3%	27.8%						
Dyspepsia								19.0%	8.0%	16.0%	1.6%
GERD								6.0%	0.0%		
Others											
Additional key AEs											
Syncope											
Dizziness								9.0%	3.0%		
Postural Dizziness	7.5%	11.1%	0.0%	8.3%	11.1%						
Somnolence / sedation						7.9%	6.7%	5.0%	4.0%		
Weight gain						2.2%	5.6%	0.0%	1.0%		
Hypertension								10.0%	1.0%	6.4%	1.6%
Insomnia										5.6%	7.8%
Akathisia								1.0%	1.0%		
Most Common TEAEs											
Headache	3.8%	22.2%	80.0%	66.7%	38.9%			14.0%	12.0%	11.2%	11.7%
Dizziness	7.5%	11.1%	0.0%	8.3%	11.1%						

Source: Company reports; FDA Label; Brannan et al. NEJM 2021; Kaul et al. Lancet 2024; Kaul et al. JAMA Psychiatry 2024

Our base case for the Ph 2 ZEPHYR study readout in 3Q26 is for ML-007C-MA to demonstrate comparable efficacy to KarXT in Ph 3 studies but be able to superior on tolerability and convenience

Scenario 1 – Base case – Superior on Tolerability and Convenience with Comparable efficacy (70% POS, +mid-to-high 20s stock move)

In our base case we assume that ML-007C-AM will demonstrate a clinical profile in its Ph 2 that reinforces the differentiation versus Cobenfy we saw in its HV studies, where it is clearly differentiated on safety with fewer AEs of vomiting, anti-cholinergic AEs (particularly urinary retention), along with at least comparable efficacy. This also assumes that it will retain its differentiation on convenience in terms of dosing (QD vs BID), food effect, and DDI.

We believe this is the most likely scenario based on all the data we have seen so far, as discussed previously. We would expect the stock to go up significantly in the mid-to-high 20s range in this scenario.

The AEs are Cmax mediated which drive the higher AE rate for the QD dose. There is a small chance that based on the AE rates management decides to move forward with the BID dose, although management may still choose to move both doses forward to maintain the optionality for patients that can tolerate the QD dose. We would still view ML-007C-MA as very meaningfully differentiated, although acknowledge that the positive stock move may be slightly smaller in magnitude.

Scenario 2 – Downside case – Comparable on both efficacy and tolerability and superior on convenience (20% POS, -low 30s stock move)

In this scenario, presumably all the tolerability advantage we have seen in HVs is not replicated in Phase 2. We believe this scenario is not very likely because while we believe we could see some anticholinergic AEs in a larger dataset, we believe ML-007C-MA should at least retain its differentiation on lower cases of vomiting, and low/no case of urinary retention.

While we did not see **anticholinergic** AEs in the doses that were taken into Phase 2, we did see these AEs in the other doses tested in HVs, the 165/3mg and the 270/6mg doses. The agonist to antagonist ratios for these two doses were the two lowest ratios tested, although the 330/6mg dose has the same 55:1 ratio as the 165/3mg dose, and so, theoretically, we could see these AEs emerge in this dose. The agonist to antagonist ratio for the 210/3mg dose is 70:1, which leans more procholinergic, and so, it is relatively less likely to have anticholinergic AEs.

With regard to the **procholinergic** AEs, we know that in general ML-007C-MA has been designed to lean more procholinergic, and so, while we would expect to see AEs such as nausea, hyperhidrosis, an investor debate is whether we would start to see some cases of vomiting in patients. This is hard to have high conviction on, although we would be surprised to see vomiting as high as 9 - 16% seen in the Cobenfy Ph 2/3 study.

Further, the emphasis on procholinergic probably also explains why we did not see AEs such as **urinary retention** in the HV study, and so we are less likely to see that emerge in a big way in patients.

If the tolerability differentiation narrows in Ph 2, a key debate is whether the convenience profile alone will be sufficient for commercial differentiation. We believe there is an unmet need in this population for a more convenient treatment with no DDI or food effect. This was corroborated in our KOL conversations who noted that dosing convenience is a big factor especially in the community setting where ~70-80% of schizophrenia patients are treated.

In this scenario we could see pressure on the stock, but our model suggests that this profile could still justify stock level in the mid-20s range.

Scenario 3 – Upside case – Superior on Tolerability, Convenience and Efficacy (10% POS, +40% Stock Move)

We believe that ML-007C-MA does not need to differentiate on efficacy to be commercially successful, and so, as long as it demonstrates an ES comparable to KarXT's Ph 3 studies (ES of 0.61 and 0.6), that would be sufficient. That being said, preclinical studies (Susmita Chatterjee et al., 2025) showed that ML-007 is 10X more potent than xanomeline (active molecule KarXT) and has demonstrated better efficacy across a range of preclinical modules of psychosis in schizophrenia including AIH, PCP-induced hyperlocomotion, and conditioned avoidance response. This could potentially translate to superior efficacy in the clinic, however, it is hard to know how these preclinical models translate in the clinic. Further, the preclinical data was for just the agonists alone, and the relative potency of the products could also change with incorporation of the PACs.

Another risk around neuropsychiatric studies is the placebo response which we saw for emraclidine in schizophrenia where it ended up failing in larger Ph 2 studies due to a higher than anticipated placebo response. While the company has implemented appropriate strategies to mitigate placebo response ([Figure 30](#)), it is unclear how that will actually pan out in clinic.

Thus, overall, we remain optimistic around our base case and see a possible scenario for superiority on efficacy on top of tolerability and convenience based on the promising preclinical efficacy, however, it is hard to put a high POS on this scenario based on the lack of clinical data and potential concerns around placebo response.

Figure 29 - Upcoming Ph 2 Data Scenarios for ML-007C-MA

			PoS (%)	Stock Move (%)
Scenario 1	Base	Superior on tolerability and convenience with comparable efficacy	65	+mid-to-high 20s
Scenario 2	Downside	Comparable on both efficacy and tolerability but superior on convenience	20	-low-30s
Scenario 3	Upside	Superior on tolerability, convenience, and efficacy	10	+40s

Source: Needham & Company Estimates

Figure 30 - Placebo response mitigation strategies



Trial Design

- Limited # of US-only sites
- Limited # of participants
- Conservative enrollment targets
- Moderate-to-severe baseline symptom requirements



Trial Execution

- Rigorous site and investigator selection
- Robust eligibility and duplicate patient controls
- Regular placebo response training of site, staff, and patients



Trial Oversight

- In-house site relationships, monitoring, and data quality oversight
- Independent central review and regular blinded data analyses
- CRO selection

Similar Strategies Employed in the Ongoing Phase 2 VISTA Study for ADP

Source: Company presentation

KEY DEBATE #2: CAN THE MUSCARINICS WORK BEYOND SCHIZOPHRENIA?; CAN ML-007 DIFFERENTIATE ITSELF IN THESE OTHER INDICATIONS?

Xanomeline has previously demonstrated benefit in AD patients on both cognition and psychosis validating potential of muscarinic agonists beyond schizophrenia

Besides schizophrenia, ML-007 is also being assessed in AD Psychosis for which the topline data is expected in 2H27; however, the more imminent catalyst is Cobenfy’s data readout which is expected by the end of 2026. There is existing preclinical data supporting the MoA in AD psychosis, but a positive pivotal trial will go a long way in further de-risking the potential of the muscarinic class.

Several other companies are also exploring indications beyond schizophrenia. Neurocrine has disclosed its muscarinic platform strategy which includes other indications such as bipolar mania, Alzheimer’s psychosis, and Lewy body dementia (LBD). Separately, BMS management has also noted expansion of its program into indications such as AD cognition, AD agitation, and bipolar disorder.

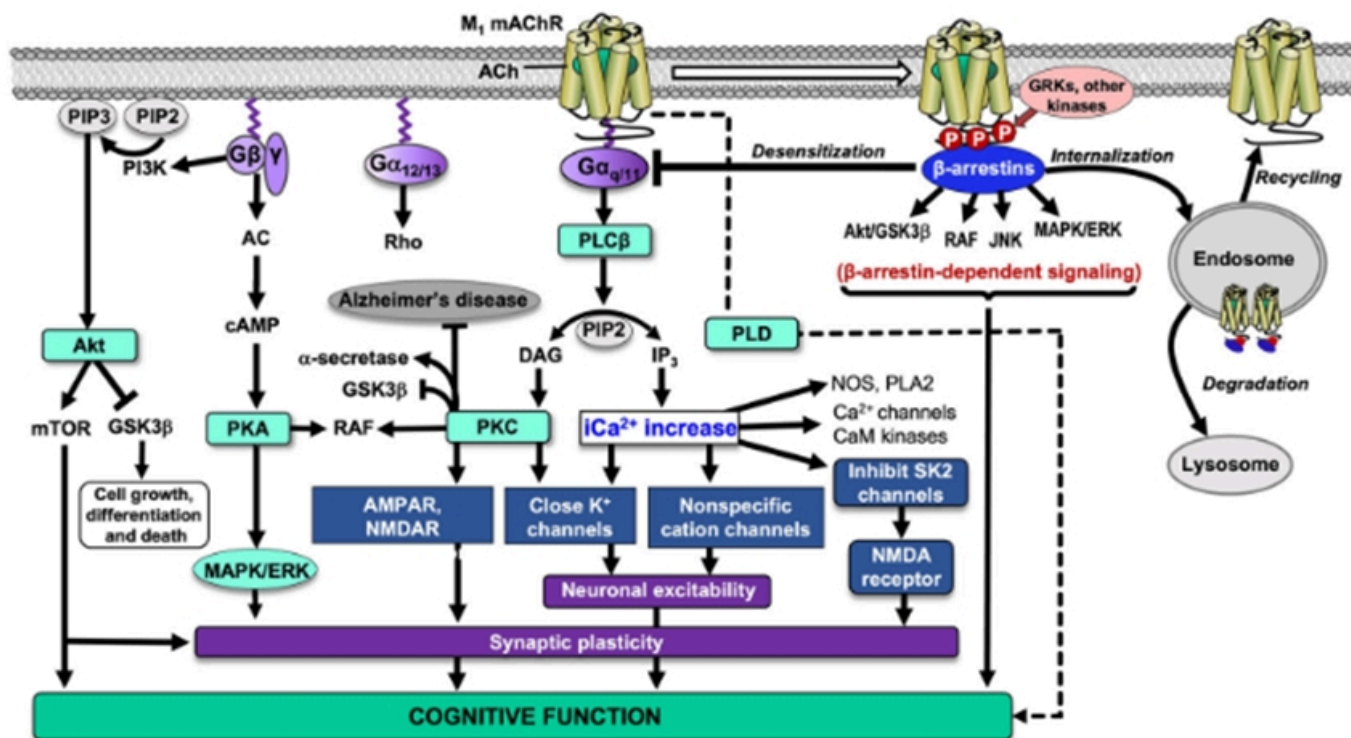
Below, we highlight the existing preclinical and clinical data that support further development in AD psychosis:

Mechanistic rationale in AD Psychosis

This report is intended for joshua.flax@westwicks.com. Unauthorized distribution prohibited.

- Acetylcholine muscarinic system has been implicated in both psychosis and cognition with anticholinergic load found to be associated with lower performance on complex attention and declarative memory task (Minzenberg et al., 2004).
- The initial rationale for using muscarinic receptor agonism was based on the cognitive impairment and psychotic symptoms observed with the administration of muscarinic antagonists such as atropine and scopolamine (Claudia Vingerhoets et al., 2017, Jan Svoboda et al., 2017)
- M1 receptors are the most abundant muscarinic subtype in the cerebral cortex, hippocampus, and prefrontal cortex which are the regions involved in cognition, memory formation, and regulation of subcortical circuits (Levey et al., 1991; Figure 31). Followed by M4 receptors which are abundantly expressed in striatum and striatal functional connectivity is known to be altered in psychosis (David Antonio Grimaldi et al., 2025)
- Thus, while AD and schizophrenia have distinct etiologies, both are said to be linked to dysfunction in the central cholinergic system with emphasis on M1/M4 receptors based on expression in the CNS.

Figure 31 - Impact of M1 on Cognitive Function



Source: Huong T M Nguyen et al., 2024

Key preclinical studies supporting development in AD psychosis

M1/M4 KO studies have demonstrated dopaminergic dysregulation and psychosis-like phenotypes:

- M1 KO (Albert A. Davis et al., 2010) mice crossed with APP_{Swe/Ind} transgenic mice demonstrated that loss of M1 AChRs decreased shedding of the neuroprotective APP_{sα} molecule in primary neurons and increased production of neurotoxic Aβ peptides, ultimately exacerbating amyloid plaque pathology in vivo.
- Another study (Caccamo et al., 2011) extended these findings and demonstrated that **M1** agonism rescued **learning and memory** impairments in transgenic mouse models of AD.
- Similarly, **M4** KO studies have demonstrated **enhanced dopaminergic stimulation** as well as **hyper-locomotion phenotype** and deficits in **startle response** (Gomez et al., 1999, Jeon et al., 2010, Dencker et al., 2011). While these studies for M4 were conducted in either WT or schizophrenia mouse models, it demonstrates M4's role in dopamine regulation and psychosis-like behavior generally

which has more of an indirect claim on the AD pathology, however, some of these preclinical findings are further supported by the clinical studies described below which target both M1/M4.

Clinical studies supporting development in AD psychosis

- Evidence in ADP was established long back in the 1990s when a placebo controlled study of xanomeline in 343 AD patients (Neil C. Bodick et al., 1997) demonstrated dose-dependent reductions in vocal outburst, suspiciousness, delusions, agitations, and hallucinations. In addition, it demonstrated improvement in cognitive function in patients with AD based on the significant treatment effect observed on ADAS-Cog (high dose vs. placebo, $p \leq 0.05$) and CIBIC+ ($p \leq 0.02$). However, in the high-dose arm, 52% of patients discontinued treatment because of adverse events; dose-dependent adverse events which were predominantly gastrointestinal in nature. The high dose group also saw Syncope in 12.6% of patients. While the development was halted in AD due to the dose-limiting gastrointestinal side-effects, this does support the efficacy of this MoA in AD psychosis based on the behavioral effects as well as cognitive benefits.
- There were also other therapies in development such as GSK1034702 (by GlaxoSmithKline) and AZD6088 (by AstraZeneca) which were both targeting M1 receptor and demonstrated promise in preclinical studies, however, did not progress beyond Ph 1 due to safety concerns.
- Overall, despite the studies failing in clinic, there is substantial evidence supporting the MoA in AD Psychosis which will be further de-risked based on the ADEPT-2 data readout later this year.
- Other companies that are also developing muscarinic receptor agonists are focusing on AD psychosis as the next indication after schizophrenia, including Neuroscine Biosciences with NBI-569 (dual M1/M4 agonist) and Abbvie with emraclidine (M4 PAM).

The delay in ADEPT-2 study readout for Cobenfy in AD psychosis does not suggest any increased risk to the potential of ML-007C-MA in AD psychosis, in our view

- In December, BMS announced the delay of ADEPT-2 Ph 3 study readout for Cobenfy in AD psychosis to 2H26 (previously end of 2025; [LINK](#)). This was based on an interim data analysis for efficacy and safety where the DMC recommended the study continue to enroll additional patients to the original target study population as the blinded review identified irregularities due to clinical trial execution at a small number of study sites.
- We believe this delay does not suggest any risk on the potential of muscarinic agonist in AD psychosis as highlighted in the previous section.
- Given that BMS remained blinded to the data and enrollment adjustments seems to be based on irregularities at the sites, it should also not have impacted the alpha as alpha allocation is only required when superiority analyses are specified at the interim look ([FDA Guidance](#))
- Overall, we see limited risk for ML-007C-MA from the delay in ADEPT-2 study readout for KarXT

ML-007C-MA can potentially be more differentiated in AD psychosis than in schizophrenia based on the unmet need in this patient population and most likely a better clinical profile compared to Cobenfy, especially on tolerability

There remains a high unmet need in AD Psychosis as there are no approved therapies in this indication. Antipsychotics are typically used off-label and tend to have severe safety and tolerability concerns, especially in an elderly population with black box warning citing increased risk of death in elderly patients. For instance, Rexulti which is an approved medication for AD agitation has a black box warning for elderly patients with dementia-related psychosis based on increased risk of death due cardiovascular or infectious issues.

KarXT is being evaluated in a Ph 3 ADEPT-2 study, wherein five different TID doses of KarXT are being evaluated - 20/2 mg (total daily dose [TDD] 60/6 mg), KarXT 30/3 mg (TDD 90/9 mg), KarXT 40/4 mg (TDD 120/12 mg), KarXT 50/5 mg (TDD 150/15 mg), KarXT 66.7/6.67 mg (TDD 200/20 mg). This was based on a Ph 1 study (KAR-030) in healthy elderly ([Figure 33](#)), where patients were titrated upto 150/20 mg, 200/30mg, and 200/40 mg (divided TID) across three cohorts. The data from Kar-030 has not been fully disclosed, but we found some data points in the FDA integrated document that can help us understand the differences in the safety profile ([LINK](#)).

- Overall, the safety in the healthy elderly patients was worse than the safety in the healthy adult patients with fewer than half of the 36 subjects being able to tolerate the planned doses for KarXT.
- There were 47% of subjects in the treatment arm who had at least one moderate AE compared to only 8% in the placebo arm whereas in the healthy adult study, at the marketed doses, the moderate AE rates were 5.7–33.3%.
- There was also one subject who had a Grade 3 urinary retention and a small intestinal obstruction which was not ruled out as treatment-related.
- Additionally, 20% of patients discontinued treatment due to AE in the treatment arm vs 0% in the placebo arm

In September 2025 Maplight announced the initiation of its Ph 2 VISTA study in AD Psychosis, which will evaluate a 210/30 mg BID dose in 300 patients ([Figure 32](#)).

We believe ML-007C-MA’s more tolerable and convenient profile will be even more relevant in this population.

As noted previously in the schizophrenia discussion, the doses we believe ML-007C-MA should be able to differentiate itself with a more tolerable profile, particularly on vomiting, and anticholinergic adverse events.

The safety profile for ML-007C-MA worsened in the healthy elderly patients compared to the healthy adults ([Figure 33](#)), which is to be expected as was seen with KarXT. However, compared to KarXT, we believe that ML-007C-MA demonstrated a more tolerable profile even in the healthy elderly subjects.

- Discontinuations:** The ML-007C-MA study had **no discontinuations**, whereas the KarXT study saw **20% discontinuation** rate. In the ML-007C-MA study only two patients reduced their dose from 210/3 mg BID (Ph 2 target dose for ADP) to 165/3mg BID.
- Moderate AE rates:** These were only **18%** for ML-007C-MA vs **47.2%** for KarXT.
- Lower concerning pro-cholinergic AEs, particularly vomiting:** The pro-cholinergic AEs were lower for ML-007C-MA compared to KarXT, particularly on nausea (27% ML-007 vs 33% KarXT) and vomiting (9% (only one case) ML-007 vs 19% KarXT). The ML-007C-MA cohort also had 27% hyperhidrosis, whereas none was reported for KarXT, however, nausea and vomiting are the more concerning AEs when it comes to tolerability.
- Lower known anti-cholinergic AEs:** From the anti-cholinergic AEs, the only comparable data point we have is on constipation which was only 9% for ML-007C-MA vs 50% for KarXT.

While we acknowledge that the doses assessed in healthy elderly subjects for KarXT were different from what is being taken into clinic. However, the 50/5 mg (**TDD 150/15 mg**) and 66.7/6.67 mg (**TDD 200/20 mg**) are close to what was used in the healthy elderly volunteers (150/20 mg and 200/30 mg). The lower amount of trospium in the ADEPT-2 study could potentially result in slightly lower rates of anticholinergic AEs, but the main AEs that result in tolerability concerns for Cobenfy in the RW are nausea and vomiting (procholinergic AEs), and the lower trospium ratio in the ADEPT-2 doses could mean relatively worse pro-cholinergic AEs. This clearly suggest that ML-007C-MA has the better safety profile even in healthy elderly subjects, which should translate similarly in the clinic.

Cobenfy TDD Tested

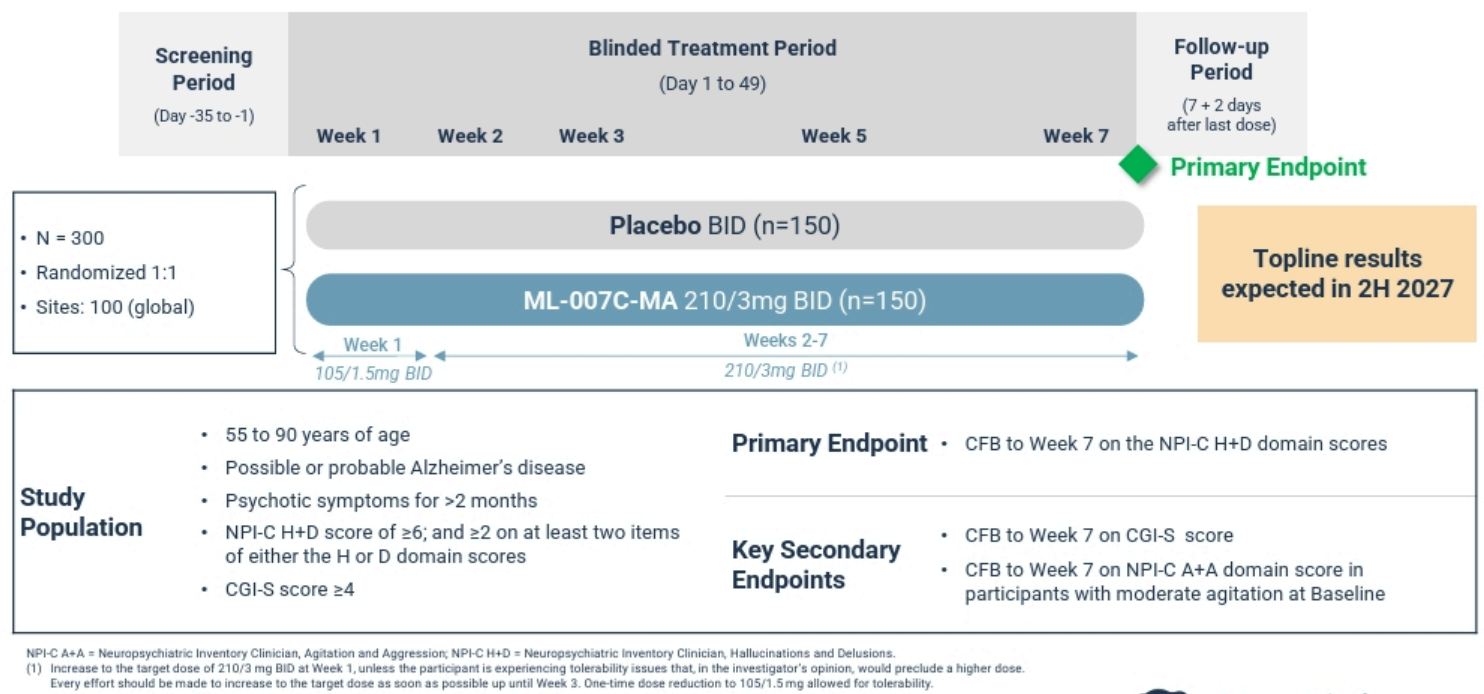
Ph 1 HV Elderly	ADEPT-2
	60/6 mg
	90/9 mg
	120/12 mg
150/20mg	150/15 mg
200/30mg	200/20 mg
200/40mg	

One consideration for ML-007C-MA relative to KarXT is the use of the PAC (festerodine), which may carry a higher potential for drug–drug interactions compared to trospium. This could be relevant in an AD population where polypharmacy is common, although the clinical impact will ultimately depend on how the formulation performs in practice.

Lastly, ML-007C-MA’s dose of BID is more convenient versus KarXT’s TID dosing, which will be an important differentiator in the patient population.

Overall, we view the safety profile for ML-007C-MA to be fairly competitive based on what we can tell by comparing the available datasets across both the healthy adults and elderly patients

Figure 32 - VISTA Ph 2 Study Design



Source: Company Presentation

Figure 33 - Healthy Elderly Volunteer Safety Comparisons - ML-007C-MA vs KarXT

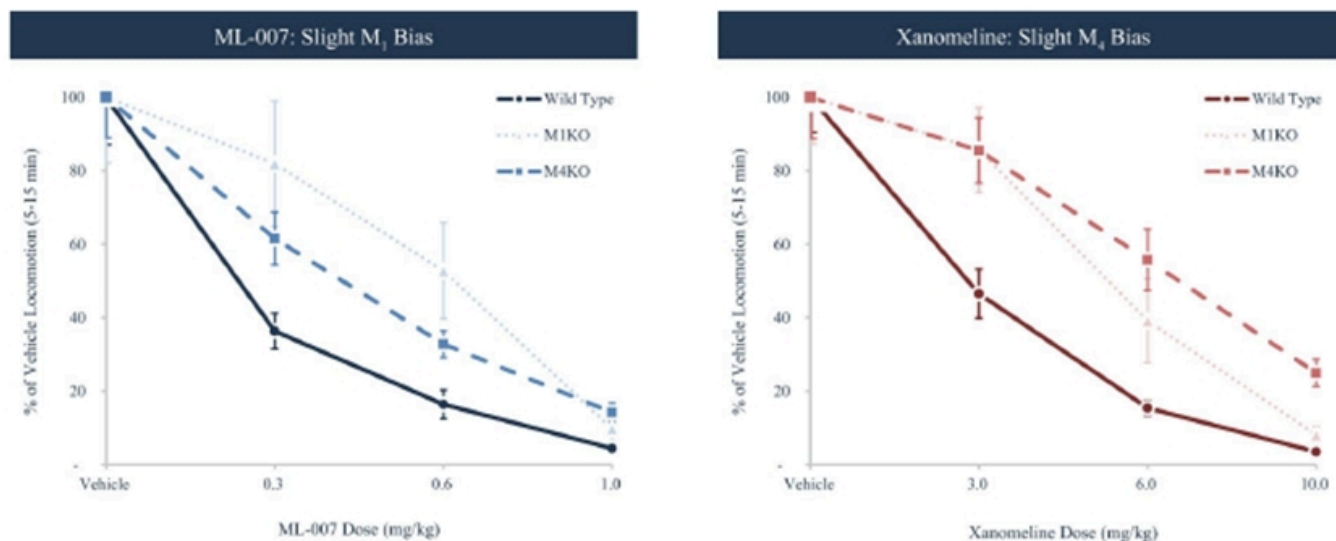
Therapy Company	ML-007 Maplight Therapeutics		Cobenfy (KarXT) BMS	
Phase	Study 013		Ph 1b	
Patient Population	Healthy Elderly		Healthy Elderly Volunteers	
Study Design	2-7 Day Titration plus 7 days at Target Dose		PCT; randomization 3:1	
Dose	Placebo combined	210 / 3mg BID (with 2-7d titration)	Placebo	KarXT
Comments on doses				150/20 mg, 200/30 mg, and 200/40 mg (divided TID)
N	7	11	12	36
SAFETY				
Any TEAE	71%	73%		
Mild	71%	73%		
Moderate	0%	18%	8.3%	47.2%
Severe	0%	0%		
AE leading to discontinuations	29%		0%	20%
Study drug-related AE leading to discontinuation		18%		
Dose reductions				
Discontinuations		Two participants discontinued study drug (placebo) due to AEs. Two participants on 210/3 mg BID dose reduced to 165/3 mg BID after experiencing AEs.		
Procholinergic	14%	55%		
Hyperhidrosis	0%	27%		
Nausea	0%	27%	0.0%	33.0%
Vomiting	0%	9%	0.0%	19.0%
Anticholinergic	14%	36%		
Dry mouth			0.0%	36.0%
Constipation		9%	17.0%	50.0%
Dyspepsia			8.0%	47.0%
Others	43%	73%		
Additional key AEs				
Syncope				
Postural Dizziness				
Most Common TEAEs				
Headache		36%		
Tremor		27%		
Hyperhidrosis		27%		
Nausea		27%		
Dizziness	14%	27%		

Source: Company Presentation; FDA Integrated Review For Cobenfy

The PK profile of ML-007C-MA as well as the Ph 2 design positions it well for success

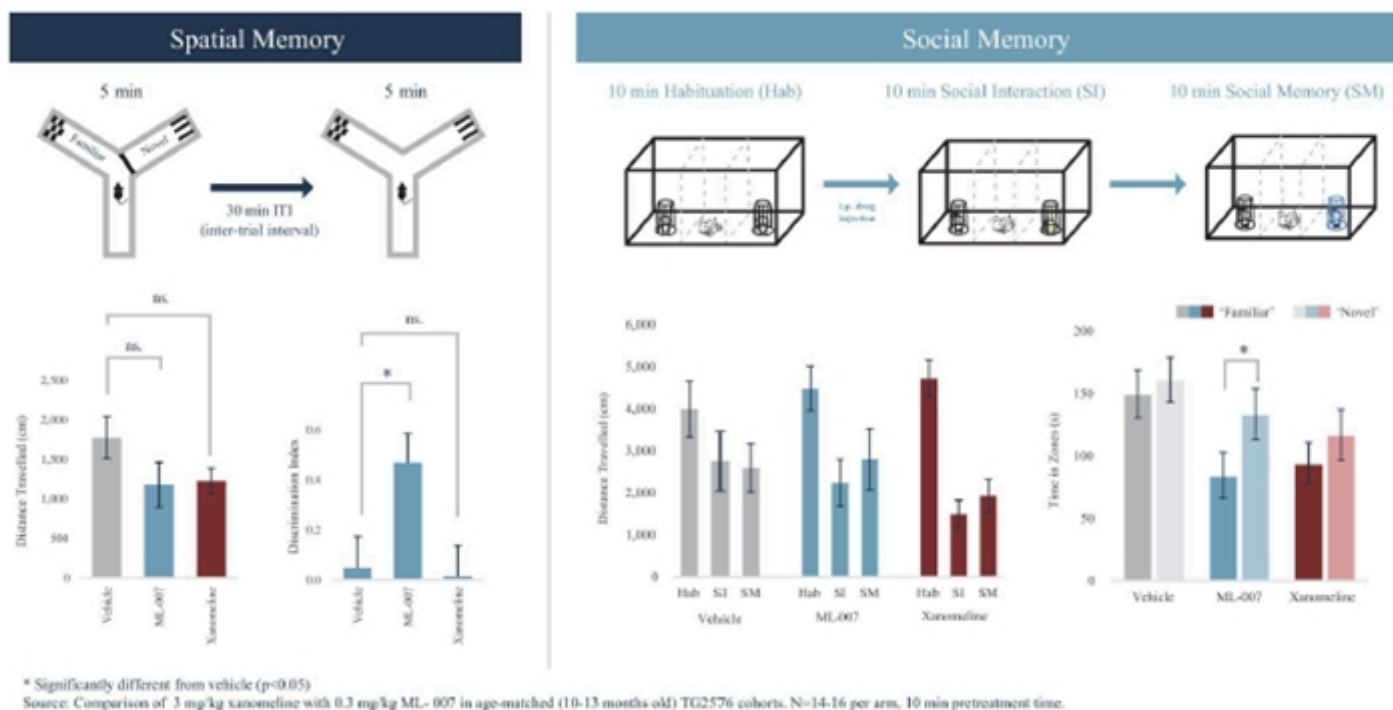
- Timepoint for measuring the primary endpoint.** While the VISTA (ML-007C-MA) study is evaluating the same primary endpoint of NPI-C:H+D, as being evaluated in KarXT's ADEPT-2, it will be measured at an earlier timepoint of 7 weeks vs 14 weeks for KarXT. This was an intentional decision by management for two reasons. First, AD psychosis is a relaxing and waning disease with a natural relapse in symptoms every 12 – 14 weeks, so it wanted to avoid that timepoint. Second, previous studies have shown a higher placebo response at 14 weeks.
- Greater M1 bias along with higher preclinical potency:** The potential for higher efficacy is supported by the design of ML-007 molecule which is ~10X more potent than xanomeline in all animal models as well as its activity that is more skewed towards M1 compared to KarXT which is more skewed towards M4 (Figure 34). The receptor bias may be even more favorable in case of AD psychosis as M1 receptors play a higher role in cognition and memory which is also validated based on the preclinical assays, namely Y-maze spatial reference memory assay (Figure 35) models unlike xanomeline

Figure 34 - M1 vs M4 Bias for ML-007 vs Xanomeline



Source: Company Reports

Figure 35 - Preclinical assays comparing spatial and social memory for ML-007 vs Xanomeline



Source: Comparison of 3 mg/kg xanomeline with 0.3 mg/kg ML-007 in age-matched (10-13 months old) TG2576 cohorts. N=14-16 per arm, 10 min pretreatment time.

Source: Company reports

Remlifsenserin (by Acadia Pharmaceuticals) is another therapy in development for AD psychosis, however, there are limited insights into how this may compare to ML-007C-MA

- Remlifsenserin (by Acadia Pharmaceuticals) is an optimized version of pimavanserin (selective 5-HT_{2A} receptor inverse agonist and antagonist) with potential to show improved efficacy over pimavanserin and reduced QT prolongation. It is being assessed in AD psychosis with the topline data readout from the Ph 2 study expected in August to October 2026.
 - Previously, pimavanserin was studied in the same population, however it was eventually not approved for AD psychosis following a negative AdCom where it was noted that the evidence around efficacy was not convincing enough ([LINK](#)).

- This version could potentially be an improvement on efficacy compared to the previous version, however, it is harder to compare either a muscarinic agonist or remlifanserin on efficacy as we are yet to see data from either therapeutic class
- Based on Acadia's management, it has suggested that remlifanserin could potentially be differentiated from Cobenfy in this patient population based on QD dosing (Cobenfy's is TID), no DDI, no food effect (Cobenfy has food effect)
- These are similar features where ML-007C-MA is differentiated, although it is taking forward a BID dose in Ph 2 VISTA study in ADP
- It is unclear whether there would be a safety differentiation between remlifanserin and ML-007C-MA as we do not have data in healthy elderly adults for either pimavanserin or remlifanserin, however, the safety data for pimavanserin in Ph 3 study for ADP was fairly clean with mostly comparable AE rates between the treatment and placebo arm (Figure 36) besides prolonged QT interval
- Currently, Nuplazid (pimavanserin) has a black box warning for elderly patients with dementia given that it is an atypical antipsychotic, however, remlifanserin is noted to be an improvement over pimavanserin with reduced QT prolongation which may or may not result in a black box warning
- We wait for the topline data for remlifanserin in August to October 2026 to make further analysis

Figure 36 - Pimavanserin Ph 3 Safety Data in AD psychosis patients

	Pimavanserin		
	AD Psychosis		
	Ph 3		
	Tx Arm (N=105)	Placebo Arm (N=112)	Tx (N=392)
	Double-blind phase		OL Phase
AE	41.00%	36.60%	36.20%
Serious AE	4.80%	4%	5.10%
AE causing discontinuation	8.60%	8.90%	7.70%
Most Common AEs			
Anxiety	2.9%	0.0%	1.5%
Asthenia	2.9%	0.9%	0.8%
Confusional state	1.0%	0.0%	2.0%
Constipation	1.9%	0.9%	2.6%
Diarrhea	0.0%	2.7%	1.3%
Dizziness	2.9%	0.0%	1.5%
Headache	9.5%	4.5%	1.5%
Hypertension	1.9%	1.8%	2.3%
Nasopharyngitis	1.0%	3.6%	1.8%
Nausea	0.0%	1.8%	2.0%
Prolonged QT interval	2.9%	0.0%	0.5%
UTI	6.7%	3.6%	5.1%
Weight decreased	1.0%	2.7%	1.3%

Source: Pierre N. Tariot et al., 2021

KEY DEBATE #3: WHAT IS THE MARKET POTENTIAL FOR THIS CLASS IN SCHIZOPHRENIA & BEYOND?

Based on KOL feedback and initial strong performance of Cobenfy, we believe that the muscarinic class can achieve ~\$5.5B in US sales in schizophrenia by 2035 with ~12% market penetration

In schizophrenia, we see a peak potential of >\$5B in US sales for the muscarinic class by 2035 with ~12% market penetration based on KOL feedback on this class and the initial strong performance of Cobenfy in this patient population

- The KOL we spoke with was a strong adopter of Cobenfy with ~22% of her patients already on treatment. This was based on the drug's efficacy which was noted to be a significant improvement over current SoC with patients becoming functional again in some cases and being able to return to work
 - The main improvement was seen on positive symptoms, however, unlike other therapies for schizophrenia, Cobenfy did not worsen the negative symptoms and improved negative symptoms in younger patients
 - Lastly, Cobenfy's safety profile was also an improvement over the current SoC with no weight gain, EPS, or sexual dysfunction
- The opportunity in schizophrenia is also supported by Cobenfy's performance in its first full launch year, where the US revenues came in at \$155M, much higher than some of the other atypical antipsychotics were able to achieve such as Caplyta which only had revenues of ~\$22.5M in its first year and ~\$82M in its second year of launch in schizophrenia

We see a peak opportunity of ~\$1.5B for ML-007C-MA in schizophrenia based on the potential to demonstrate an improvement in tolerability profile and convenient dosing

ML-007C-MA can potentially achieve ~\$1.5B in US sales by 2035 in schizophrenia based on the potential to demonstrate an improvement in the safety profile along with convenient dosing regimen.

- The initial data comparisons between preclinical and data in healthy volunteer demonstrate that ML-007C-MA can potentially achieve a better tolerability profile in schizophrenia patients compared to Cobenfy with no vomiting and no anti-cholinergic AEs seen in healthy volunteer adults with the Ph 2 target dose of 210/3mg BID and 330/6 mg QD
- In terms of dosing, as stated before, there is potential for optionality with both QD and BID dosing.
- In addition, there is minimal titration and no fasting requirements whereas titration with Cobenfy's low dose sometimes can take up to weeks in the real world setting and the food requirements can be intensive impacting compliance or resulting in GI side-effects
- Overall, we see ML-007C-MA being able to achieve ~\$1.5B in US sales by 2035 with a peak penetration of ~30% by 2035 within the muscarinic class given that it is likely to be able third in terms of launch timing behind Cobenfy and direclidine (by Neurocrine Biosciences)

This was supported by our KOL conversations and RW data which highlights the limitations of Cobenfy's tolerability profile

- Majority of patients (60-65%) on Cobenfy in her clinic developed GI side effects such as nausea and vomiting and required prescription of additional antiemetic medications.
- None of the patients were unable to titrate up to the highest dose of 125/30 mg BID on the label due to these tolerability issues
- This seems to be in-line with the RW data presented by BMS (at Psych Congress, September 2025) which noted that ~64.4% and 36.7% of patients experienced nausea and vomiting, respectively, in the RW setting (N=90). Additionally, within the patient population in the RW analysis, only 17.8% of patients were able to be on the max dose of 125/30 mg BID whereas 56.7% stayed at 100/20 mg BID.
- The KOL also noted that the BID dosing for Cobenfy can have compliance issues especially in the community setting (~70-80% of schizophrenia patients are treated in this setting) based on limited caregiver support.

In AD psychosis, we see a higher potential for ML-007C-MA with potential peak revenues of ~\$2.4B based on lack of approved therapies and greater unmet need in this patient population.

- In AD psychosis, for muscarinic class, we see a higher peak penetration of ~50% by 2035 compared to schizophrenia and a peak potential of >\$7.5B based on the lack of approved therapies in this patient population and higher unmet need with the use of antipsychotics as noted in the section above.
- For ML-007C-MA, we see a peak U.S. opportunity of ~\$2.4B and ~22% market penetration by 2035 based on the lack of approved therapies and potential to meaningfully differentiate from competition in AD psychosis.
 - There are limited late-stage competitors in development with BMS disclosing ADEPT-2 topline data readout this year and ACAD with Ph 2 data readout for ACP-204.
 - The ADEPT-2 study readout for Cobenfy should actually be positive for ML-007C-MA as it de-risks the entire class in AD psychosis population.
 - Based on the safety data in healthy elderly volunteers and preclinical data, ML-007CMA can potentially demonstrate a better clinical profile compared to Cobenfy (elaborated in Debate 2)
- During our conversation with the KOL, she noted that she sees AD psychosis with a bigger unmet need given the lack of treatment options and sees the muscarinic class taking a higher share in this patient population.

At present, we assign a PoS of 55% for ML-007C-MA in schizophrenia based on the initial promising preclinical and healthy volunteer data and a lower PoS of 25% in AD psychosis due to lack of confirmatory studies for this class.

- We assign ML-007C-MA a PoS of 55% in schizophrenia based on the proof of muscarinic class being applicable and viable in this patient population, promising preclinical data which suggests higher potency for ML-007 compared to KarXT, direclidine, and emraclidine, and safety data in healthy adult volunteers which suggest an improvement in tolerability profile compared to Cobenfy.
- For AD psychosis, we assign a lower PoS of 25% for ML-007C-MA based on the lack of confirmatory studies with this class, although if the ADEPT-2 study is positive, that would change how we view this opportunity.
 - In December, BMS announced the delay of ADEPT-2 Ph 3 study readout for Cobenfy in AD psychosis to 2H26 (previously end of 2025; [LINK](#)). This was based on an interim data analysis for efficacy and safety where the DMC recommended the study continue to enroll additional patients to the original target study population as the blinded review identified irregularities due to clinical trial execution at a few study sites. We believe this delay does not suggest any risk on the potential of muscarinic agonist in AD psychosis.

Larger opportunity for muscarinic class as development expands beyond schizophrenia

- There is potential for muscarinic receptor agonism to work in therapeutic indications beyond schizophrenia and AD psychosis. The muscarinic acetylcholine receptors consist of 5 distinct subtypes and have a central role in human physiology including many fundamental functions of the CNS (Andrew C. Kruse et al., 2014). Based on preclinical work in mAChR-mutant mice, it has been observed that targeting mAChRs can be beneficial in Alzheimer's disease, drug addiction, type 2 diabetes, and cancer (Andrew C. Kruse et al., 2014).
- In addition, we are seeing more proof from a clinical perspective based on the development strategy employed by different companies for the muscarinic class such as with Cobenfy which is being assessed in multiple indications as part of its LCM strategy, including AD psychosis, AD agitation, AD cognition, bipolar-I mania (adj. & monotherapy), and pediatric autism irritability
- Similarly, NBIX held an R&D day where it disclosed its development strategy for muscarinic class in other indications such as bipolar mania, Alzheimer's cognition, & LBD
- Maplight itself recently disclosed a new asset, ML-021, a M4 antagonist in its pipeline which is being developed for Parkinson's Disease.

- Overall, there is potential for a much larger TAM for the muscarinic class which benefits MPLT as it demonstrates success with each muscarinic asset and indication.

SCHIZOPHRENIA

US	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E
Injectable/IV Sub-total (Month, Year)	2,812,111	2,783,990	2,826,612	2,869,887	2,913,825	2,958,435	3,003,728	3,049,714	3,096,405	3,143,810	3,191,942
Injectable/IV Sub-total (Quarter)											
% Growth YOY	-4.2%	-1.0%	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%
% Share of Total Category	3.9%	3.8%	3.9%	3.9%	3.9%	4.0%	4.0%	4.0%	4.1%	4.1%	4.2%
Orals Sub-total (Month, Year)	69,556,741	69,904,525	70,165,746	70,324,529	70,437,014	70,455,891	70,488,866	70,648,213	70,863,365	71,140,645	71,454,100
Orals Sub-total (Quarter)											
% Growth YOY	-1.0%	0.5%	0.4%	0.2%	0.2%	0.0%	0.0%	0.2%	0.3%	0.4%	0.4%
% Share of Total Category	96.0%	95.9%	95.7%	95.4%	95.0%	94.4%	93.9%	93.6%	93.4%	93.2%	93.0%
Muscaranics Sub-Total (Month, Year)	98,468	200,489	320,782	545,330	817,995	1,186,093	1,541,920	1,773,208	1,950,529	2,067,561	2,150,263
% Growth YOY	2361.7%	103.6%	60.0%	70.0%	50.0%	45.0%	30.0%	15.0%	10.0%	6.0%	4.0%
% Seq Growth Monthly	14%										
Muscaranics Sub-Total (Quarter)											
% Growth YOY	2362%	104%	60.0%	70.0%	50.0%	45.0%	30.0%	15.0%	10.0%	6.0%	4.0%
% Share of Total Category	0.1%	0.3%	0.4%	0.7%	1.1%	1.6%	2.1%	2.3%	2.6%	2.7%	2.8%
Total Category TRxs	72,467,320	72,889,004	73,313,141	73,739,746	74,168,834	74,600,418	75,034,514	75,471,136	75,910,299	76,352,017	76,796,305
% Growth YOY	-1.0%	0.6%	0.6%	0.6%	0.6%	0.6%	0.6%	0.6%	0.6%	0.6%	0.6%

TRx - Muscaranics											
Cobenfy	98,468	200,489	320,782	518,063	736,195	937,013	967,555	834,516	688,476	547,338	426,924
Direclidine			0	0	65,440	154,192	285,255	401,188	501,652	579,725	640,333
ML-007			0	0	16,360	94,887	227,433	351,317	460,508	547,015	614,820
NMRA - 861 / NMRA - 898			0	0	0	0	30,838	115,259	192,615	256,507	307,588
Muscarinic Sub-Total (Month, Year)	98468	200,489	320,782	545,330	817,995	1,186,093	1,541,920	1,773,208	1,950,529	2,067,561	2,150,263

Market Share - Muscaranics											
Cobenfy	100%	100%	100%	95%	90%	79%	63%	47%	35%	26%	20%
Direclidine					8%	13%	19%	23%	26%	28%	30%
ML-007					2%	8%	15%	20%	24%	26%	29%
NMRA - 861 / NMRA - 898							2%	7%	10%	12%	14%
Others							2%	4%	6%	7%	7%
Muscarinic Sub-Total	100%	100%	100%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%

ML-007	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E
TRxs			0	16,360	94,887	227,433	351,317	460,508	547,015	614,820
% Growth Rate - YOY										
% Growth Rate - QOQ										
Capture rate				85%	85%	85%	85%	85%	85%	85%
Adjusted TRx				19247	111632	267569	413314	541774	643547	723318
Price per Rx - Gross			\$1,898.4	\$2,001.3	\$2,109.7	\$2,224.0	\$2,344.4	\$2,471.3	\$2,605.1	\$2,746.0
Price per Rx - Avg Monthly Gross										
Gross to Net Adjustment			25.0%	25.5%	26.0%	26.5%	27.0%	27.5%	28.0%	28.5%
Price per Rx - Net			\$1,518.7	\$1,594.6	\$1,674.4	\$1,758.1	\$1,846.0	\$1,938.3	\$2,035.2	\$2,137.0
% Growth Rate - YOY			5%	5%	5%	5%	5%	5%	5%	5%
Inventory										
Sales			\$0.00	\$30.7	\$186.9	\$470.4	\$763.0	\$1,050.1	\$1,309.8	\$1,545.7
% Growth Rate - YOY			#DIV/0!	#DIV/0!	509%	152%	62%	38%	25%	18%
VA Cons.			\$0.0	\$29.3	\$90.2	\$202.3	\$316.8	\$434.9	\$570.6	\$702.8

AD PSYCHOSIS

US	2024	2025	2026	2027	2028	2029	2030	2031E	2032E	2033E	2034E	2035E
Alzheimer Disease Patients	6,900,000	7,200,000	7,408,800	7,623,655	7,844,741	8,072,239	8,306,334	8,547,217	8,795,087	9,050,144	9,312,598	9,582,664
% Growth Rate	2.9%	2.9%	2.9%	2.9%	2.9%	2.9%	2.9%	2.9%	2.9%	2.9%	2.9%	2.9%
Diagnosis rate	36.0%	36.9%	37.7%	38.5%	39.2%	39.9%	40.5%	41.1%	41.6%	42.2%	42.7%	43.1%
AD Patients	2,484,000	2,655,000	2,792,770	2,932,390	3,073,988	3,217,695	3,363,643	3,511,965	3,662,795	3,816,270	3,972,527	4,131,705
% AD Patients that have Psychosis	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%
AD Psychosis Pts	993,600	1,062,000	1,117,108	1,172,956	1,229,595	1,287,078	1,345,457	1,404,786	1,465,118	1,526,508	1,589,011	1,652,682
% AD Psychosis Pts with Moderate-Severe Disease	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%
No. of AD Psychosis Pts with Moderate-Severe Disease	496,800	531,000	558,554	586,478	614,798	643,539	672,729	702,393	732,559	763,254	794,505	826,341
% AD Patients Treated for Psychosis	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%
No. of AD Pts Treated for Psychosis	447,120	477,900	502,699	527,830	553,318	579,185	605,456	632,154	659,303	686,929	715,055	743,707
ACP-204 Market Share					1.4%	4.0%	6.8%	8.8%	10.4%	11.5%	12.4%	13.0%
KarXT / Cobenry Market Share				1.4%	4.0%	8.0%	11.0%	13.3%	14.9%	16.2%	17.2%	17.9%
ML-007 Market Share							1.3%	4.0%	10.5%	15.4%	19.0%	21.8%
NBI-569 Market Share								1.3%	5.0%	6.7%	8.1%	9.2%
Others				99%	95%	88%	81%	73%	59%	50%	43%	38%
Muscarinic Mkt Share				1.4%	4.0%	8.0%	12.3%	18.5%	30.4%	38.2%	44.2%	48.9%
Total Patients on ACP-204				0	7,470	23,167	40,868	55,709	68,300	79,131	88,593	96,996
Total Patients on Cobenry				7,126	22,133	46,335	66,600	83,760	98,483	111,304	122,649	132,858
Total Patients on ML-007				0	0	0	7,568	25,286	69,227	105,615	136,084	161,931
Total Patients on NBI-569				0	0	0	0	7,902	32,965	45,795	57,602	68,517
Others				520,704	523,715	509,683	490,419	459,497	390,328	345,083	310,128	283,405
ML-007												
Gross Price per Month	\$1,887	\$1,925	\$1,963	\$2,002	\$2,043	\$2,083	\$2,125	\$2,168	\$2,211	\$2,255	\$2,300	\$2,346
Price Increase	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%
Gross-to-Net	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%
Net Price per Month	\$1,415	\$1,444	\$1,472	\$1,502	\$1,532	\$1,563	\$1,594	\$1,626	\$1,658	\$1,691	\$1,725	\$1,760
Avg No. of Months on Treatment	8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4
Net yearly cost	\$11,888	\$12,126	\$12,368	\$12,616	\$12,868	\$13,125	\$13,388	\$13,656	\$13,929	\$14,207	\$14,492	\$14,781
Sales (\$M)	0.0	0.0	0.0	0.0	0.0	0.0	101.3	345.3	964.2	1,500.5	1,972.1	2,393.6
VA Cons.				\$0.0	\$0.0	\$0.0	\$31.3	\$72.9	\$205.4	\$388.6	\$592.3	\$786.4

THE SECOND ASSET, ML-004, PRESENTS A POTENTIAL UPSIDE WITH PH 2 POC DATA EXPECTED IN 3Q26, ALTHOUGH AT PRESENT IT DOES NOT IMPACT OUR VALUATION

Background

Autism Spectrum Disorder (ASD) is a highly heritable and heterogenous neurodevelopmental disorder that affects ~62M people globally (Damian F Santomauro et al., 2025, Catherine Lord et al., 2020) and ~7M in the US. The defining feature of ASD is deficits in social communication and interaction which is not fully addressed by current pharmacotherapy options (Risperdal approved for irritability associated with ASD and Abilify approved for irritability associated with ASD). In addition, these are antipsychotics that have safety concerns such as increased cardiac, GI, and nervous system disorders (FDA label). The other treatment approaches are dominated by behavioral interventions, such as Applied Behavior Analysis (ABA), speech therapy, and occupational therapy, that are limited in availability, expensive, and variable in terms of efficacy.

Previously, there have been attempts made to develop drugs that target ASD core symptoms such as Roche's balovaptan (V1a receptor antagonist) which demonstrated promising Ph 2 signals, however, ultimately failed in the Ph 3 in both children and adults with no stat difference versus placebo on the PE (Suma Jacob et al., 2022, Eric Hollander et al., 2022)

Mechanistic Rationale for ML-004

Maplight's second asset, ML004, is being developed for the treatment of social communication deficits and irritability in adolescents and adults (ages 12 to 45 years) with ASD. It is currently in Ph 2 IRIS study which is enrolling ~160 patients across the US, Australia, and Canada, and the topline data is expected in 3Q26.

ML-004 is a selective 5HT 1B/1D receptor agonist that has been formulated for QD dosing. It is an IR/ER formulation of zolmitriptan that was originally developed for migraine. The reformulation is designed for chronic use at higher doses than previously approved and could potentially increase sociability, with the aim of achieving pro-social effects seen with MDMA and psilocybin as 5-HT1B activation has been linked to increase sociability. While all three act on the serotonin system, which regulates mood, social behavior, and emotional processing, the specific receptor subtypes differ significantly, with psilocybin being a 5TH-2A receptor agonist and MDMA has a broader effect on serotonin.

This emerged from work aimed at identifying circuit biology responsible for pro-social effects seen with MDMA and psilocybin (Coris Heifets et al., 2020, Sixtine Fleury et al., 2025, Maplight's S1).

Preclinical validation for ML-004's MoA Across Multiple ASD Models

Prior to initiating clinical studies, mgt conducted in vivo animal studies with zolmitriptan to understand its therapeutic potential in treating sociability and irritability in AD. The key findings from these preclinical studies were:

- In VPA (mouse valproic acid), a model of autism, administration of zolmitriptan at a dose of 10 mg/kg significantly reversed the social deficit in these mice.
- Similarly, in CTNAP2 KO mouse line, a second model of autism, treatment with zolmitriptan 10 mg/kg significantly improved the social deficit phenotype.
- There were 5 in vivo animal studies done, including evaluation of effects on locomotion as well as to evaluate effect on zolmitriptan on sociability and aggression in ASD, all of which demonstrated improvements in sociability and aggression.

The data for Ph 2 IRIS study is expected in 3Q26 and presents an upside call option not currently part of any investor estimates

- The first Ph 1 study (MAP-ZOL-HV-001) evaluated high-dose IR zolmitriptan oral formulation in a total of 40 healthy adult volunteers randomized into five ascending dose study drug cohorts of 5 mg, 10 mg, 20 mg, 30 mg and 40 mg, or placebo dosed TID.
 - The safety and PK profile from this study supported further development. The most common AEs were headaches, dizziness, nausea, and hiccups with no clinically significant or dose-responsive changes in mean QTc.
- In the second Ph 1 study (ML-004-ER-001), ML-004 was evaluated in a total of 12 healthy adult volunteers under fasted and fed conditions.
 - The study demonstrated that a single dose of ML-004 (24 mg) resulted in prolonged absorption phase and reduced Cmax compared to IR zolmitriptan.
 - Based on these results, modeled data suggested that ML-004 48 mg and 72 mg would result in target plasma exposure based on preclinical efficacy models for approximately 12–15 hours, respectively.
 - Lastly, both zolmitriptan IR and ML-004 24 mg were generally well tolerated, and AEs were less common with ML-004 dosing than with IR zolmitriptan dosing.
- Overall, both the Ph 1 studies supported further development in Ph 2 IRIS study ([Figure 37](#)) which is a PCT expected to enroll ~160 adolescent (age 12-17) and adult (age 18-45) participants with ASD to evaluate the efficacy of ML-004 compared with placebo in the improvement of core social communication deficits.

Figure 37 - Ph 2 IRIS Study Design

- Multi-center, randomized, double-blind, placebo-controlled trial
- **Randomized approximately 160 subjects with ASD**
 - Adolescent (age 12-17) and adult (age 18-45)
- Flexible dosing paradigm, with target maintenance dose of 48mg and 72mg
- **Primary Endpoint:**
 - ABI, Social Communication Domain Score
 - Change vs. baseline assessed after 12 weeks of maintenance dosing
- **Secondary Endpoint:**
 - Change in ABC-I score vs. baseline for patients with moderate or greater irritability score at baseline
 - Other secondary and exploratory endpoints
- **Topline results expected in Q3 2026**

Source: Company Presentation

MapLight Therapeutics
Income Statement
(\$ Millions, except EPS)

	2025A	1Q26E	2Q26E	3Q26E	4Q26E	2026E	2027E	2028E	2029E	2030E
Total Revenues	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	16.9	128.1
Consensus	0.0	0.0	0.0	0.0	0.0	0.0	0.0			
COGS	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.3	9.6
Gross Profit	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	15.6	118.5
SG&A	30.7	13.0	14.1	15.0	15.2	57.3	59.0	81.6	116.5	139.9
Consensus	30.7	8.5	9.0	9.7	10.2	37.4	42.1			
R&D	138.3	42.0	40.5	37.1	39.6	159.1	165.4	182.0	164.9	152.8
Consensus	138.3	35.1	36.8	38.1	40.8	150.7				
Acquired in-process research and development	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Expenses	169.1	55.0	54.6	52.1	54.8	216.4	224.4	263.7	281.4	292.7
Operating Income	-169.1	-55.0	-54.6	-52.1	-54.8	-216.4	-224.4	-263.7	-265.8	-174.2
Non-Operating Income	7.9	2.5	2.5	2.5	2.5	10.0	10.0	10.3	10.6	10.9
Pretax Income	-161.2	-52.5	-52.1	-49.6	-52.3	-206.4	-214.4	-253.4	-255.2	-163.3
Taxes	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tax Rate	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Net loss	-161.2	-52.5	-52.1	-49.6	-52.3	-206.4	-214.4	-253.4	-255.2	-163.3
Shares - Weighted average	8.7	32.6	33.1	33.6	47.6	36.8	56.6	58.6	66.9	68.9
EPS - Basic & diluted	-\$18.56	-\$1.61	-\$1.57	-\$1.47	-\$1.10	-\$5.75	-\$3.79	-\$4.32	-\$3.81	-\$2.37
Consensus	-\$18.56	-\$0.95	-\$0.99	-\$0.98	-\$1.03	-\$3.95	-\$4.10			

Source: Company reports; Needham & Company, LLC estimates

Margin Analysis	2025A	1Q26E	2Q26E	3Q26E	4Q26E	2026E	2027E	2028E	2029E	2030E
Revenues						100.0%	100.0%	100.0%	100.0%	100.0%
COGS						NM	NM	NM	7.5%	7.5%
Gross Margin						NM	NM	NM	92.5%	92.5%
SG&A						NM	NM	NM	690.4%	109.2%
R&D						NM	NM	NM	976.6%	119.2%
Operating Margin						NM	NM	NM	-1574.6%	-135.9%
Pretax Margin						NM	NM	NM	-1511.7%	-127.4%
Net Margin						NM	NM	NM	-1511.7%	-127.4%

% Change	2025A	1Q26E	2Q26E	3Q26E	4Q26E	2026E	2027E	2028E	2029E	2030E
Revenues							NM	NM	NM	659.1%
COGS							NM	NM	NM	659.1%
Gross Profit							NM	NM	NM	659.1%
SG&A							3.0%	38.3%	42.7%	20.1%
R&D							3.9%	10.1%	-9.4%	-7.3%
Operating Income							3.7%	17.5%	0.8%	-34.5%
Pretax Income							3.9%	18.2%	0.7%	-36.0%
Net Income							3.9%	18.2%	0.7%	-36.0%

Source: Company reports; Needham & Company, LLC estimates

MapLight Therapeutics - Revenue
(\$ Millions, except EPS)

	2025A	1Q26E	2Q26E	3Q26E	4Q26E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E
ML-007C-MA															
Schizophrenia								0.0	30.7	186.9	470.4	763.0	1,050.1	1,309.8	1,545.7
VA Cons. (Unadj)								0.0	29.3	90.2	196.0	316.8	452.4	590.4	724.9
Schizophrenia PoS Adj.								0.0	16.9	102.8	258.7	419.6	577.6	720.4	850.1
VA Cons. (Adj)								0.0	17.6	53.2	135.9	216.2	308.2	398.5	485.9
Alzheimer's Disease Psychosis								0.0	0.0	101.3	345.3	964.2	1,500.5	1,972.1	2,393.6
VA Cons. (Unadj)								0.0	0.0	31.3	72.9	205.4	388.6	592.3	786.4
AD Psychosis Pos Adj.								0.0	0.0	25.3	86.3	241.1	375.1	493.0	598.4
VA Cons. (Adj)								0.0	0.0	15.7	40.7	87.8	152.2	226.1	298.9
ML-007C-MA (M1 / M4 agonist) - Total								30.7	288.2	815.7	1,727.2	2,550.6	3,281.8	3,939.3	
PoS (%)								55.0%	44.5%	42.3%	38.3%	37.4%	37.0%	36.8%	
ML-007C-MA (M1 / M4 agonist) PoS Adj.								16.9	128.1	345.1	660.7	952.7	1,213.4	1,448.5	
Growth								659.1%	169.3%	91.5%	44.2%	27.4%	19.4%		
Consensus						0.0	0.0	0.0	17.6	65.0	182.0	304.0	445.6	610.2	775.6
ML-004 (5-HT 1B/ID agonist)															
Autism Spectrum Disorder											59.0	205.4	332.9	444.8	543.9
ASD PoS Adj.											3.0	10.3	16.6	22.2	27.2
Consensus						0.0	0.0	0.0	0.0	0.0	11.6	24.2	38.4	52.7	67.2
Collaboration Revenue						0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Royalties/Collaboration Revenues						0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Revenues						0.0	0.0	0.0	16.9	128.1	348.0	671.0	969.3	1,235.6	1,475.7
Consensus						0.0	0.0	0.0	17.6	75.1	259.3	515.2	809.3	1,099.1	1,392.2

Source: Company reports; Needham & Company, LLC estimates

Maplight Therapeutics
Balance Sheet

(\$ Millions, except per-share data)

	2024A	2025A	2026E	2027E	2028E	2029E	2030E
ASSETS							
Current assets:							
Cash and cash equivalents	38.3	46.7	382.1	511.0	352.2	350.1	226.2
Short-term investments	70.5	258.4	158.4	58.4	8.4	0.0	0.0
Accounts receivable	0.0	0.0	0.0	0.0	0.0	1.2	9.0
Prepaid expenses and other current assets	5.8	19.6	20.1	20.6	21.1	21.6	22.1
Total current assets	114.6	324.7	560.6	590.0	381.6	372.8	257.2
Long-term investments	11.4	148.0	98.0	98.0	98.0	98.0	98.0
Restricted cash	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Operating lease, right-of-use asset	6.4	5.5	5.5	5.5	5.5	5.5	5.5
Property and equipment, net	1.2	1.1	1.0	0.9	0.8	0.7	0.6
Other assets	3.2	0.1	0.1	0.1	0.1	0.1	0.1
Total assets	136.9	479.5	665.3	694.6	486.2	477.2	361.5
LIABILITIES AND STOCKHOLDERS' EQUITY							
Current liabilities:							
Accounts payable	1.9	3.3	4.5	6.0	7.5	9.0	10.5
Accrued expenses and other current liabilities	10.0	12.0	14.0	16.0	18.0	20.0	22.0
Lease liability - current	0.8	0.9	1.0	1.1	1.2	1.3	1.4
Deferred grant earnings	3.2	0.0	0.0	0.0	0.0	0.0	0.0
Total current liabilities	15.9	16.2	19.5	23.1	26.7	30.3	33.9
Operating lease liability, net of current portion	5.8	4.9	4.7	4.5	4.3	4.1	3.9
Total liabilities	21.7	21.1	24.3	27.7	31.1	34.5	37.9
Stockholders' equity:							
Series D	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Series C	225.0	0.0	0.0	0.0	0.0	0.0	0.0
Series B-1	12.0	0.0	0.0	0.0	0.0	0.0	0.0
Series B	51.1	0.0	0.0	0.0	0.0	0.0	0.0
Series A-1	16.0	0.0	0.0	0.0	0.0	0.0	0.0
Series A-1	4.8	0.0	0.0	0.0	0.0	0.0	0.0
Common stock	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Additional paid-in capital	5.6	819.1	1208.2	1,448.5	1,490.0	1,732.9	1,777.0
Accumulated other comprehensive loss	0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Accumulated deficit	-199.4	-360.5	-566.9	-781.3	-1,034.7	-1,289.9	-1,453.1
Total stockholders' equity	115.2	458.4	641.1	667.0	455.1	442.8	323.7
Total liabilities and stockholders' equity	136.9	479.5	665.3	694.6	486.2	477.2	361.5

Source: Company reports; Needham & Company, LLC estimates

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Maplight Therapeutics
Cash Flow

(\$ Millions, except per-share data)

	2024A	2025A	2026E	2027E	2028E	2029E	2030E
CASH FLOWS FROM OPERATING ACTIVITIES:							
Net loss	-77.6	-161.2	-206.4	-214.4	-253.4	-255.2	-163.3
Adjustments to reconcile net loss to net cash used in operating activities:							
Depreciation	0.7	0.6	0.6	0.6	0.6	0.6	0.6
Stock-based compensation expense	1.1	36.0	40.4	41.6	42.9	44.1	45.5
Net amortization of premiums and accretion of discounts on investments	-1.7	-2.1	-2.1	-2.1	-2.1	-2.1	-2.1
Common stock issued to NeuroSolis	0.0	0.5	0.5	0.5	0.5	0.5	0.5
Non-cash lease expense	0.7	0.9	0.1	0.1	0.1	0.1	0.1
Loss from equity method investment	1.0	0.0	0.0	0.0	0.0	0.0	0.0
Loss from disposal and write-off of equipment	0.1	0.0	0.0	0.0	0.0	0.0	0.0
Changes in operating assets and liabilities:	-3.1	-12.8	2.5	2.8	2.8	1.6	-5.0
Net cash used in operating activities	-78.8	-138.1	-164.4	-170.9	-208.6	-210.3	-123.7
CASH FLOWS FROM INVESTING ACTIVITIES:							
Purchases of short-term and long-term investments	-125.0	-408.2	100.0	100.0	50.0	8.4	0.0
Maturities of marketable debt securities	45.0	85.5	50.0	0.0	0.0	0.0	0.0
Purchases of property and equipment	-0.8	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Net cash provided by (used in) investing activities	-80.8	-323.2	149.5	99.5	49.5	7.9	-0.5
CASH FLOWS FROM FINANCING ACTIVITIES:							
Payments of deferred financing costs	-2.2	-5.0	0.0	0.0	0.0	0.0	0.0
Proceeds from issuance of Series C	120.0	0.0	0.0	0.0	0.0	0.0	0.0
Proceeds from issuance of Series D	0.0	198.8	0.0	0.0	0.0	0.0	0.0
Proceeds from exercise of stock options	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Proceeds from issuance of common stock upon stock option exercises	0.0	275.7	350.0	200.0	0.0	200.0	0.0
Proceeds from long-term debt	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Net cash provided by financing activities	118.1	469.8	350.3	200.3	0.3	200.3	0.3
Increase (decrease) in cash, cash equivalents, and restricted cash	-41.5	8.5	335.4	128.9	-158.8	-2.1	-123.9
Cash, cash equivalents, and restricted cash, beginning of period	80.0	38.5	46.9	382.3	511.2	352.4	350.3
Cash, cash equivalents, and restricted cash, end of period	38.5	46.9	382.3	511.2	352.4	350.3	226.4

Source: Company reports; Needham & Company, LLC estimates

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VALUATION (PRICE TARGET: \$37.00)

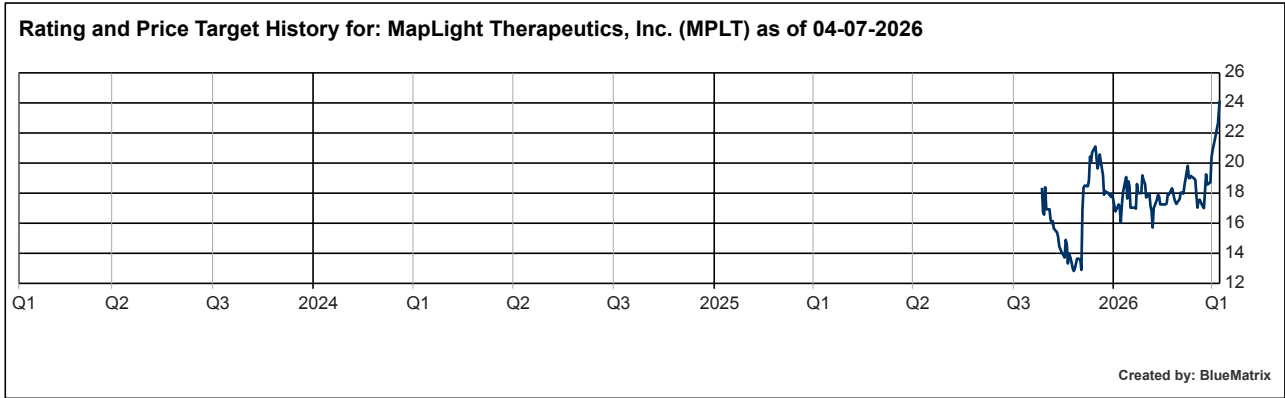
We derive our PT using a DCF valuation which uses cash flow projections out to 2035 and a discount rate of 12.5%.

POTENTIAL UPSIDE DRIVERS

- Better efficacy compared to Cobenfy drives higher market share in Schizophrenia
- ML-007C-MA has higher penetration in AD psychosis based on the unmet need and a better than expected clinical profile for muscarinics in elderly population
- Early pipeline gets negligible value currently, if successful could drive shareholder value

RISKS TO TARGET

- MPLT
- The safety profile worsens in schizophrenia patients with ML-007C-MA losing safety differentiation from Cobenfy which results in lower market penetration
 - Muscarinic agonists do not work as well in AD psychosis



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Hold	23	3
Underperform	< 1	0
Rating Suspended	0	0
Restricted	0	0

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